Computers, brains and AI

4: Deep Learning

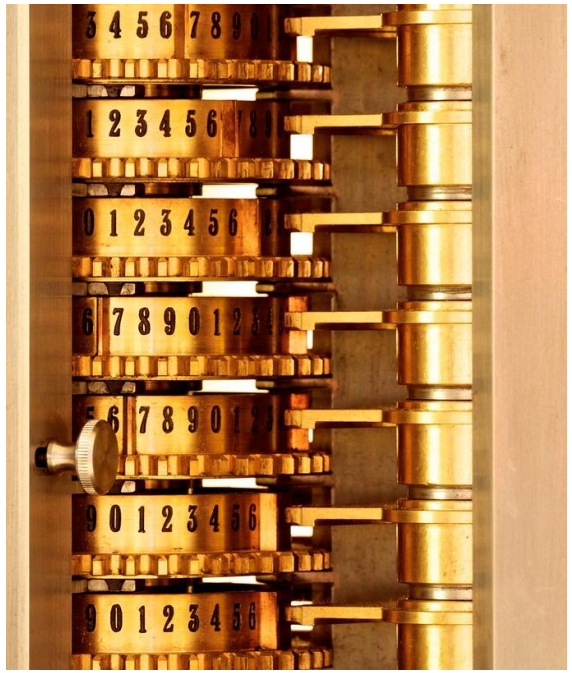




Photo by Marcin Wichary / CC BY
Photo by Science Museum Group / CC0

1) Intro

2) Foundations
3) Silicon

→ 4) Deep learning — Brains and neural networks
5) Deep learning — How?

6) ChatGPT
7) ChatGPT
8) ChatGPT

9) The future of AI
10) The future of AI

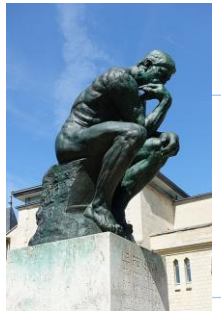


1960

1980

2000

2020



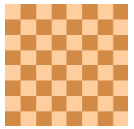
GOFAI

ELIZA

chatbot

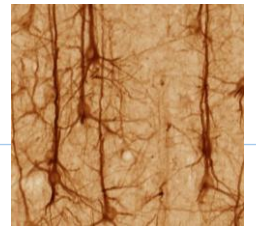
Good old fashioned AI

Easy is hard



Deep Blue

Machine learning



Perceptron

Birds fly







AlexNet



2012

Chess
Jeopardy
Book / CC0
Swan from pxhere / CC0
Bus by Martin49 / CC BY 2.0
Cat by Von.grzanka / CC BY-SA 3.0
Dog by Kyfishfishfish / CC BY-SA 4.0
The thinker by CrisNYCa / CC BY-SA 4.0
Tractor by Kharkiv Tractor Plant / CC BY 4.0
Penguin by Christopher Michel / CC BY 2.0
Neurons by UC Regents Davis campus / CC BY 3.0
Duckling by Luis Miguel Bugallo Sánchez / CC BY-SA 3.0

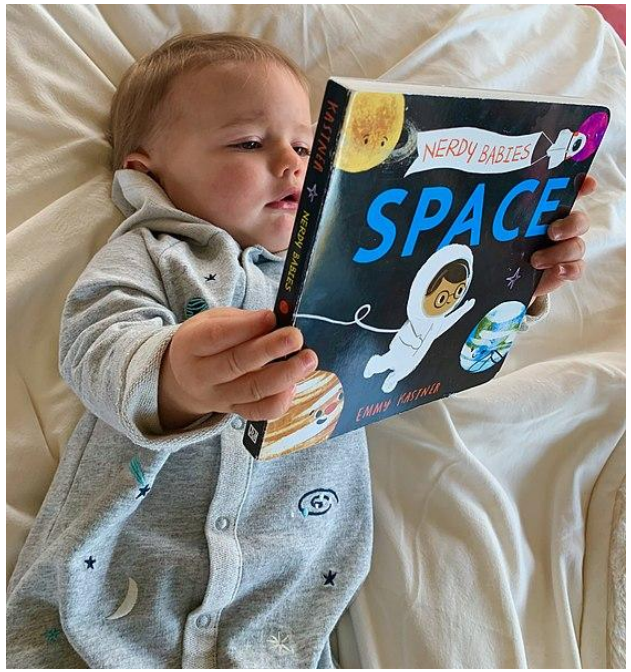
Fei-Fei Li



ImageNet (2009)

14 million ***labelled*** images

Free



ImageNet Challenge

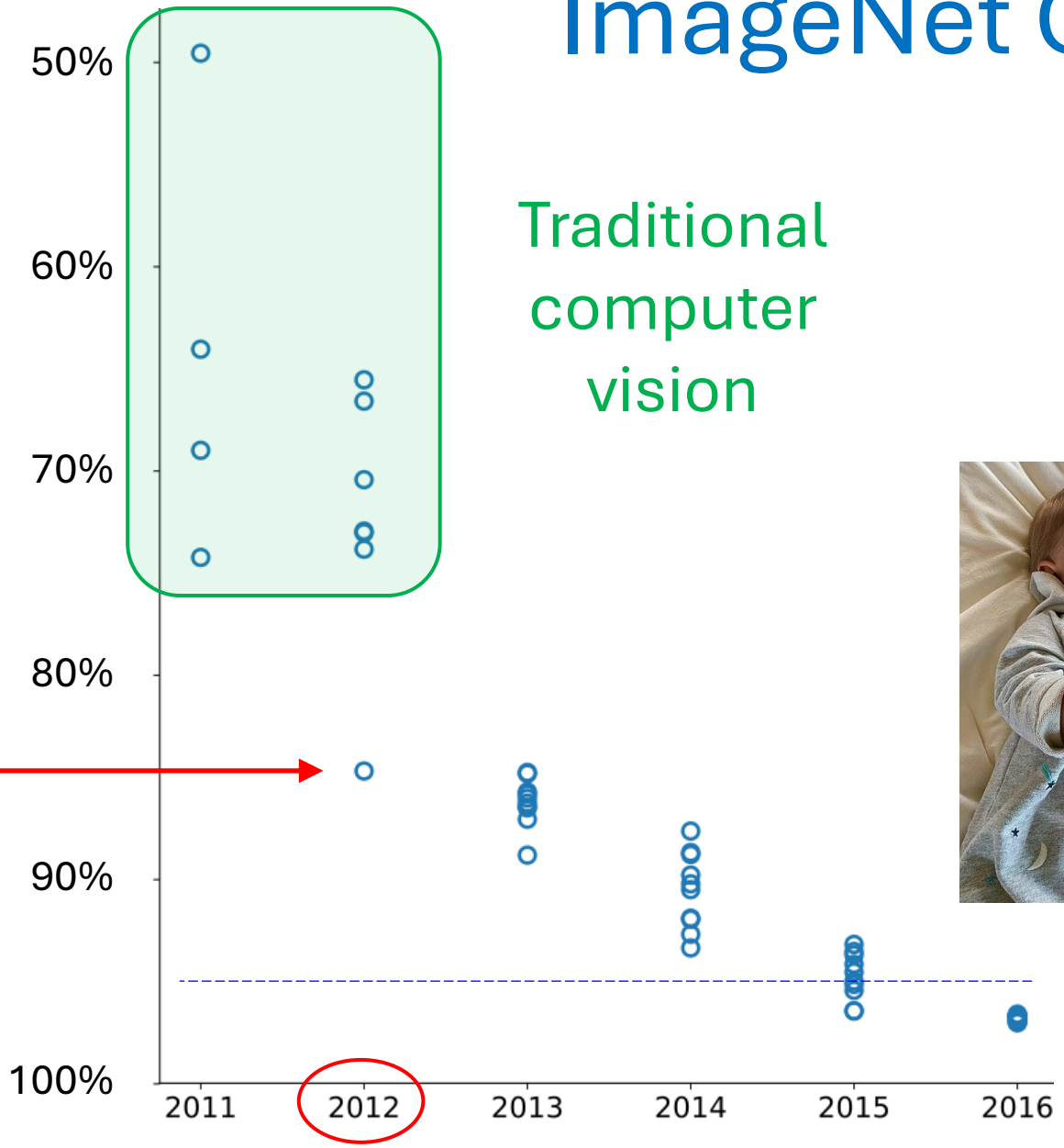
Classifying images

https://www.ted.com/talks/fei_fei_li_how_we_re_teaching_computers_to_understand_pictures

Baby by Steve Jurvetson / CC BY 2.0
Fei-Fei Li by ITU Pictures / CC BY 2.0

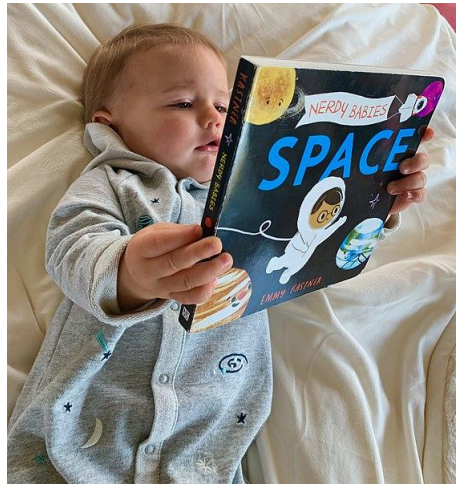
ImageNet Challenge





Traditional computer vision

AlexNet



Human level

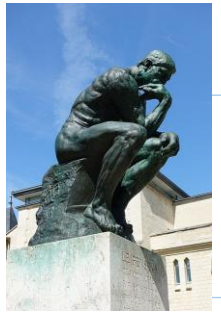
Zebra from Pixabay
Baby by Steve Jurvetson / CC BY 2.0
ImageNet by Gkrusze / CC BY-SA 4.0

1960

1980

2000

2020



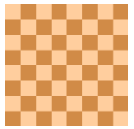
GOFAI

ELIZA

chatbot

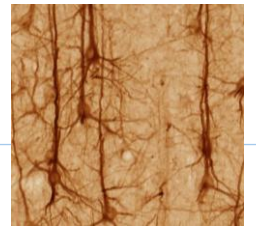
Good old fashioned AI

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AlexNet



2012

Chess
Jeopardy
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Swan from pxhere / CC0
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Penguin by Christopher Michel / CC BY 2.0
Neurons by UC Regents Davis campus / CC BY 3.0
Duckling by Luis Miguel Bugallo Sánchez / CC BY-SA 3.0

AlexNet (2012)



bus



dog



cat

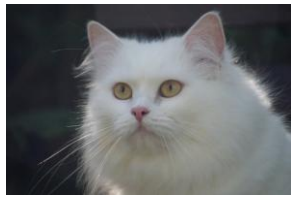


tractor



cat

• • •



cat



AlexNet (2012)

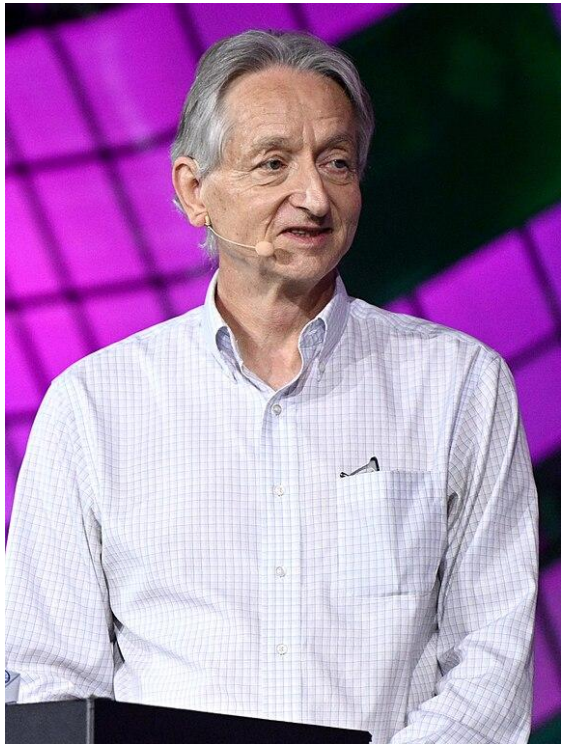
157,000 citations

Alex Krizhevsky

Ilya Sutskever



Geoffrey Hinton



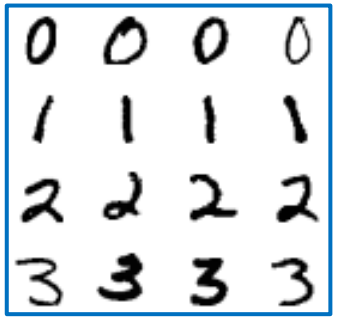
Hinton By Ramsey Cardy / CC BY 2.0
Sutskever by Eladkarmel / CC BY-SA 4.0



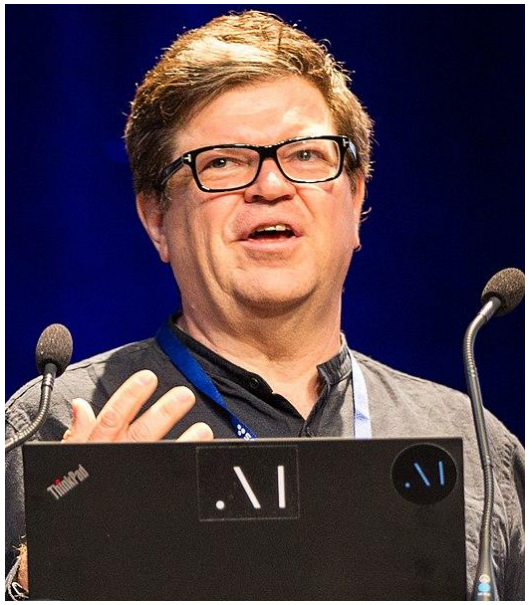


Photo by Oren Jack Turner / CC0

		Layers
LeNet	1988	5
DanNet	2011	?
AlexNet	2012	8
ResNet	2015	152



Yann LeCun

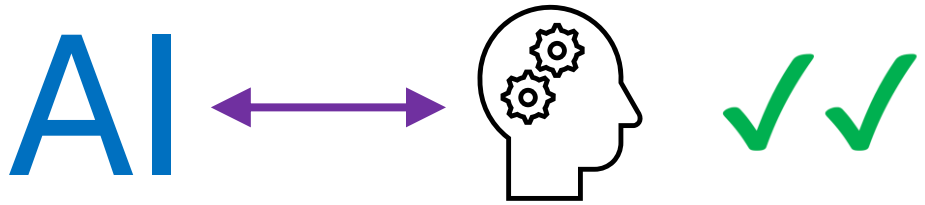


AI risk sceptic



brains and AI

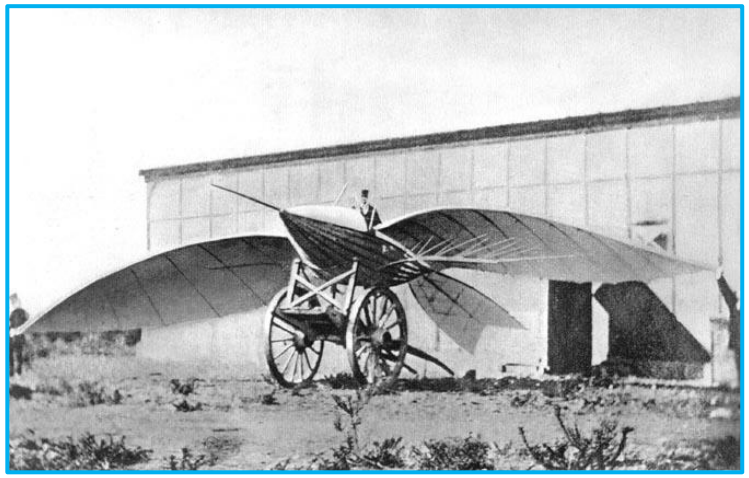
MNIST by Suvanjanprasai / CC BY-SA 4.0
Yann LeCun By Jérémy Barande / CC BY-SA 2.0

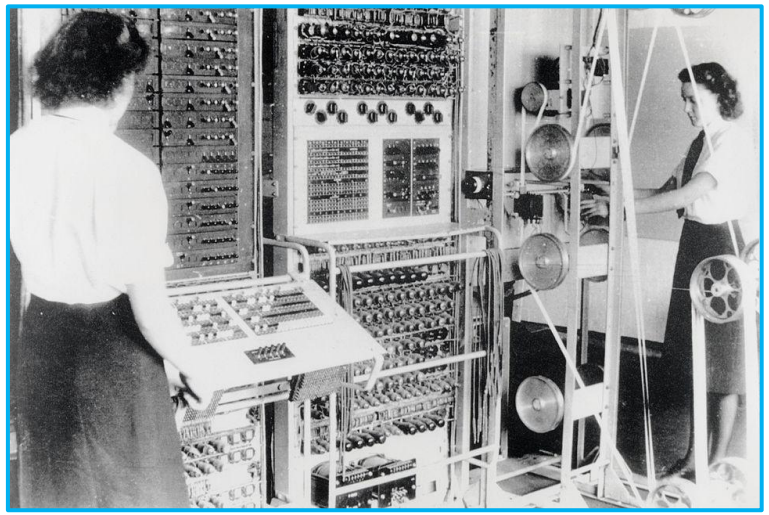

AI



brain

✓



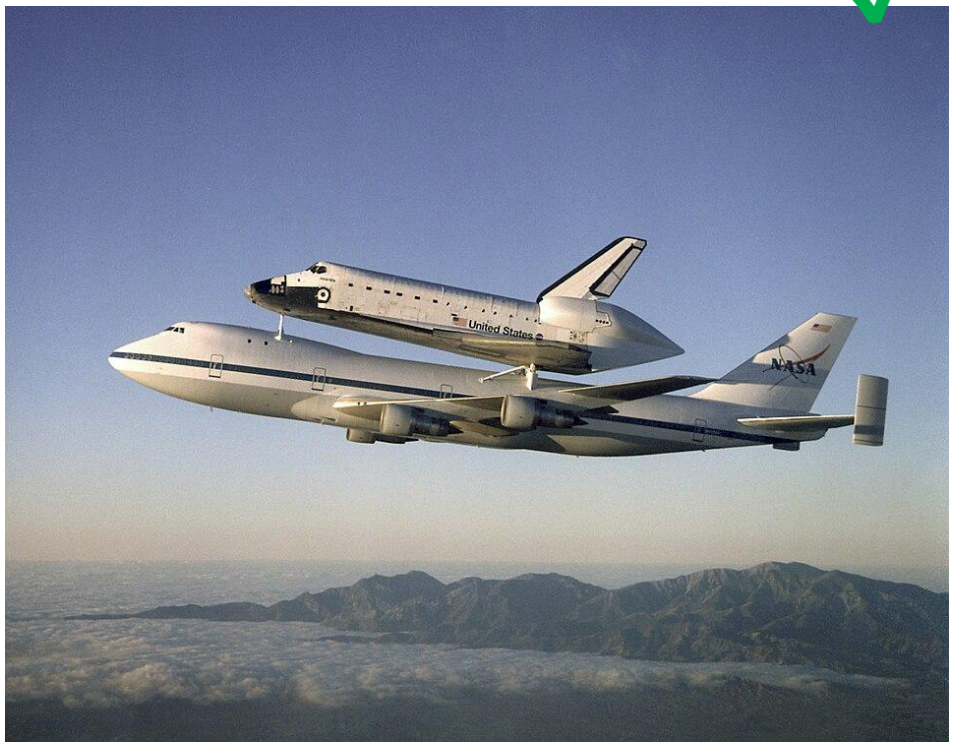


✗

computer ✗

artificial neural network ✓

✓

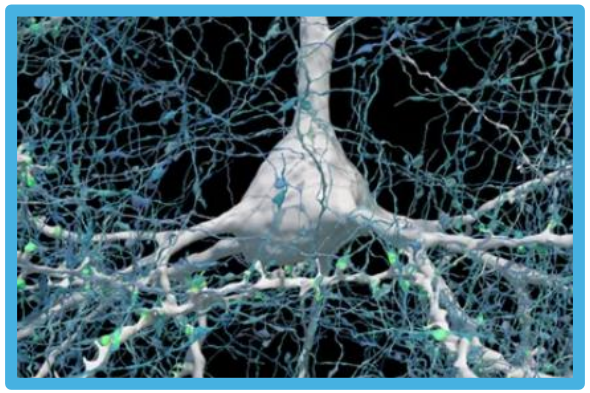


💡

Colossus / CC0
Owl by Richard Lee / Unsplash
Glider by Pépin fils from Brest / CC0
Jumbo By NASA / Carla Thomas / CC0



https://www.nature.com/articles/d41586-024-01387-9



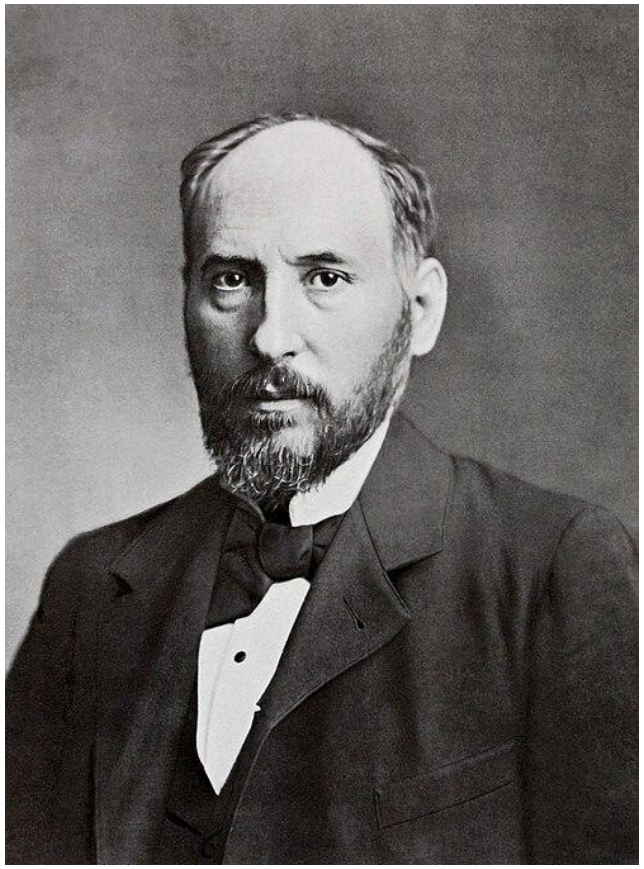
https://blog.google/technology/research/google-ai-research-new-images-human-brain/

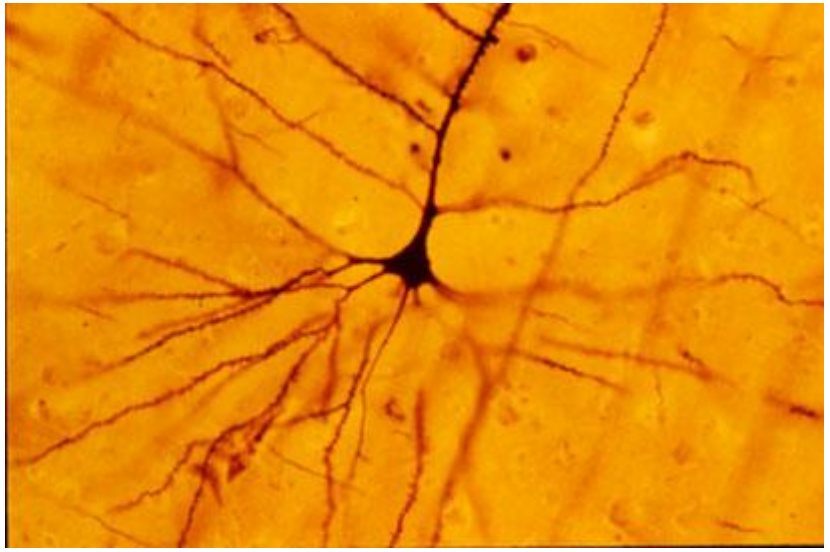
https://h01-release.storage.googleapis.com/gallery.html

Camillo Golgi



Santiago Ramón y Cajal



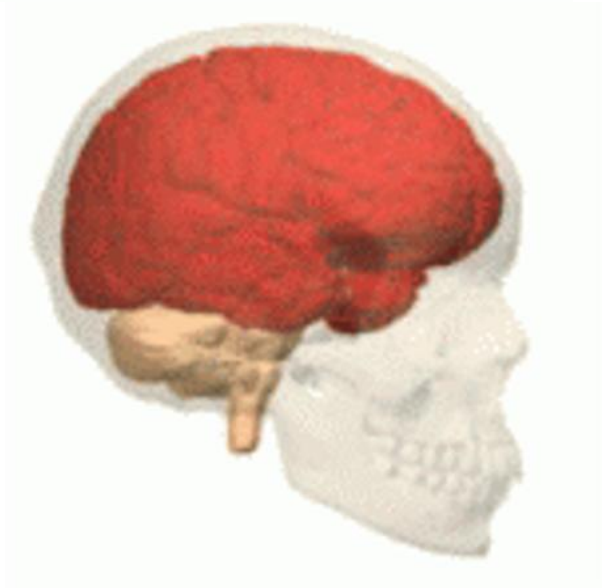


Golgi's method
1873

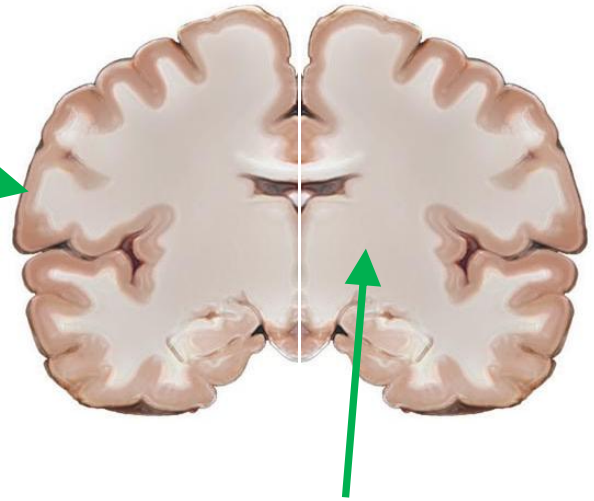
Nobel Prize - 1906

https://youtu.be/F64cnuo0oNU?t=3

Neuron by Bob Jacobs / CC0
Golgi by unknown author / CC0
Cajal by Clark University in 1899, restoration by Garrondo / CC0



Neocortex

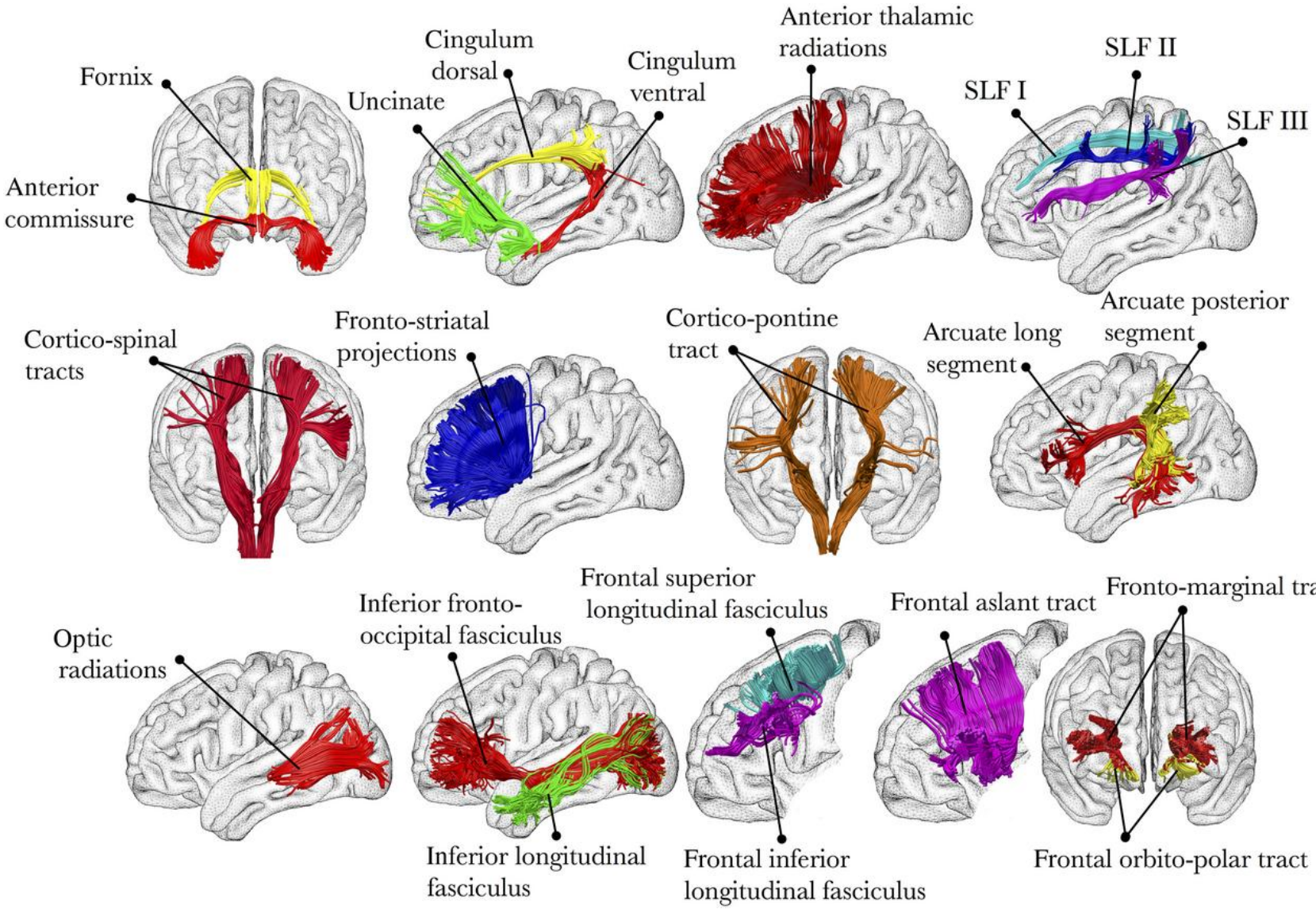


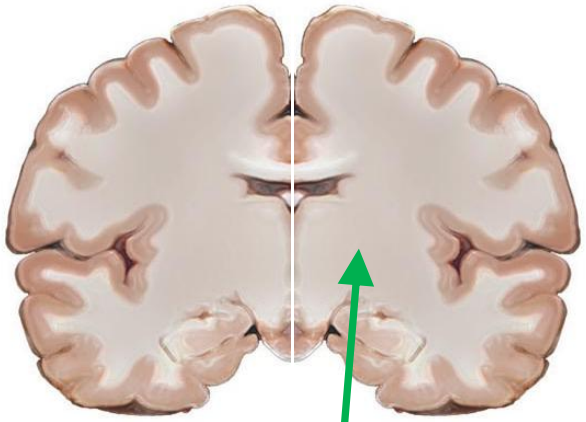
White matter

Long-range connections

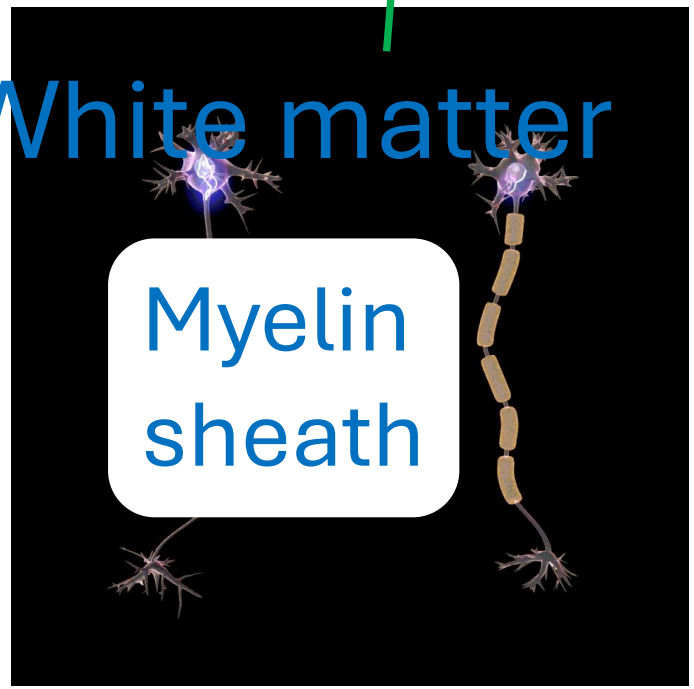
tea towel

Brain by National Institutes of Health / CC0
Cerebrum by Database Center for Life Science / CC BY-SA 2.1 jp

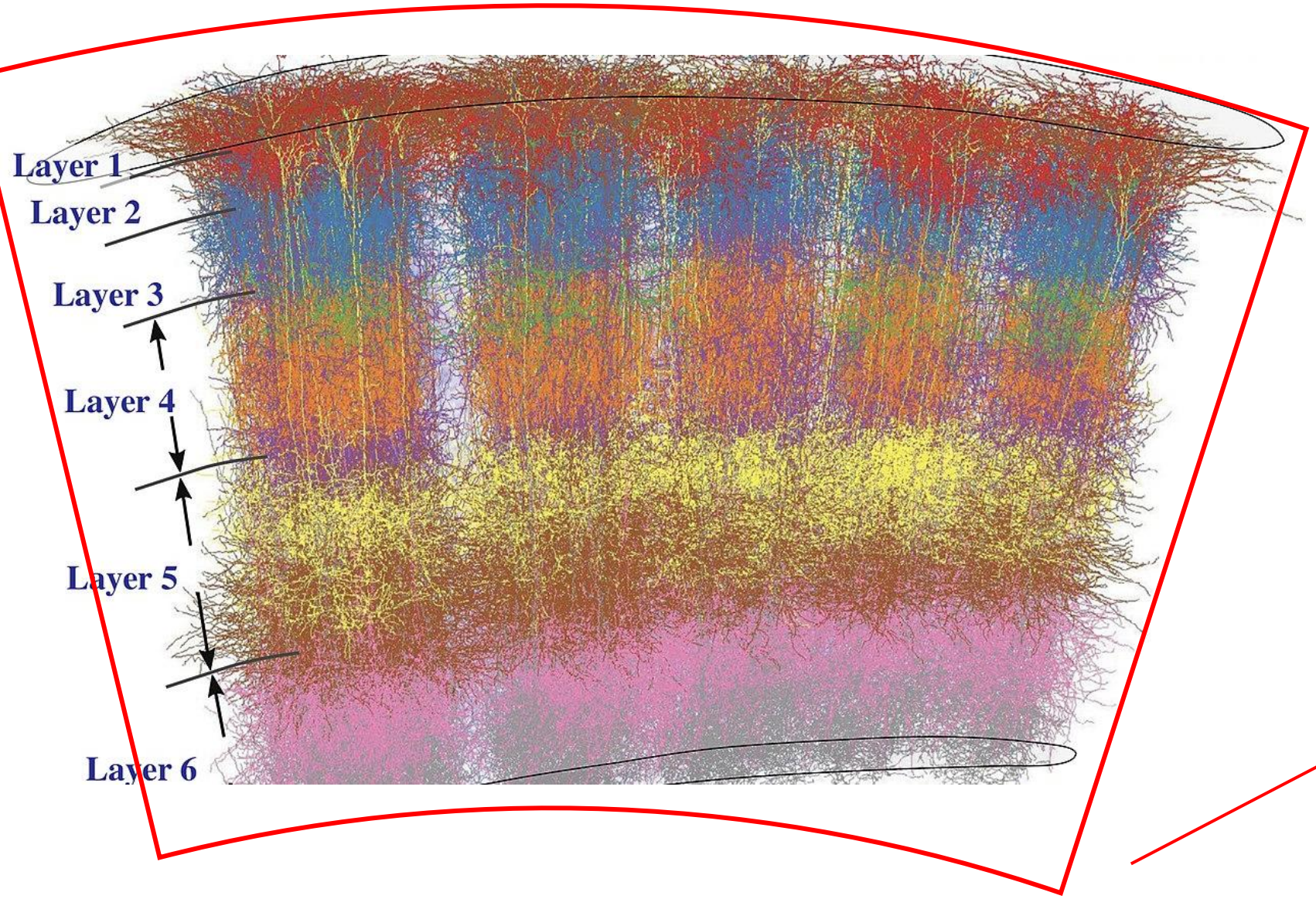


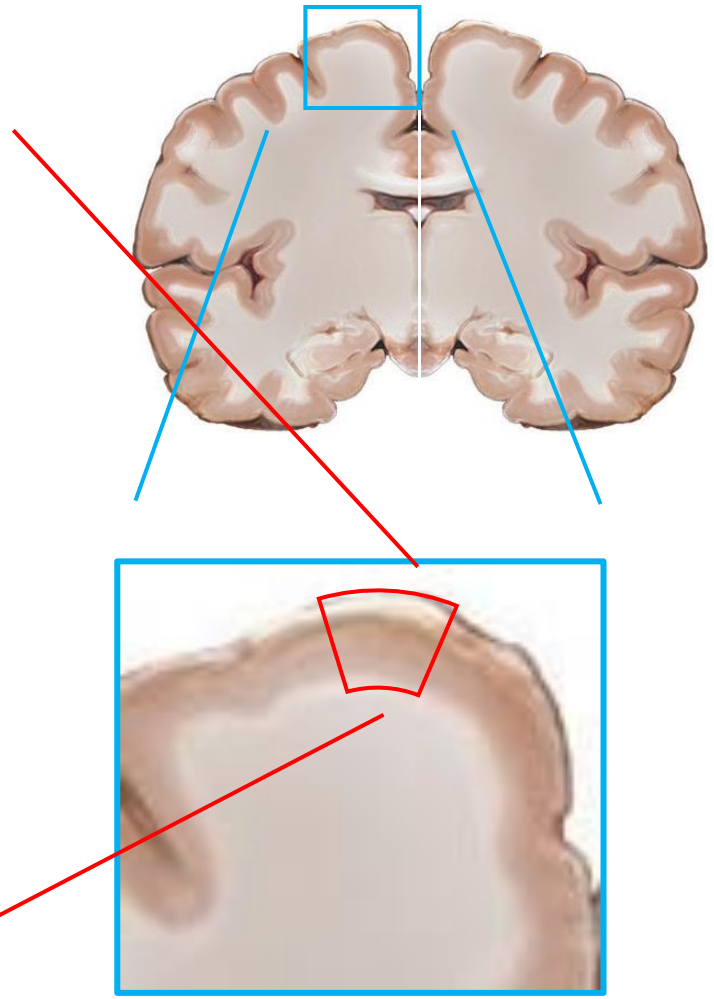


White matter



Myelin by Dr. Jana / CC BY 4.0
Brain by National Institutes of Health / CC0
Tracts by Thiebaut de Schotten M, Dell'Acqua F, Ratiu P, et al. / CC BY 4.0

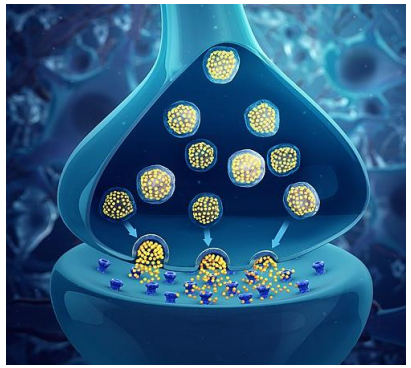


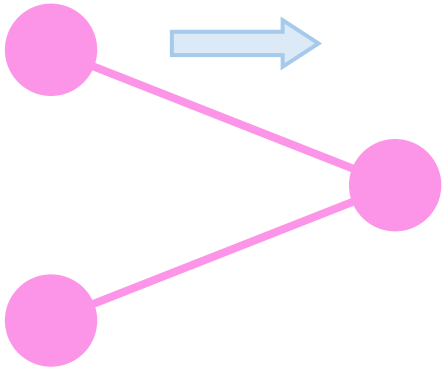


Thousand brains theory How similar brains are

Neocortex by Bodysurfinyon / CC BY-SA 4.0
Myelin by Dr. Jana / CC BY 4.0
Tracts by Thiebaut de Schotten M, Dell'Acqua F, Ratiu P, et al. / CC BY 4.0
Brain by National Institutes of Health / CC0
Cerebrum by Database Center for Life Science / CC BY-SA 2.1 jp

synapse

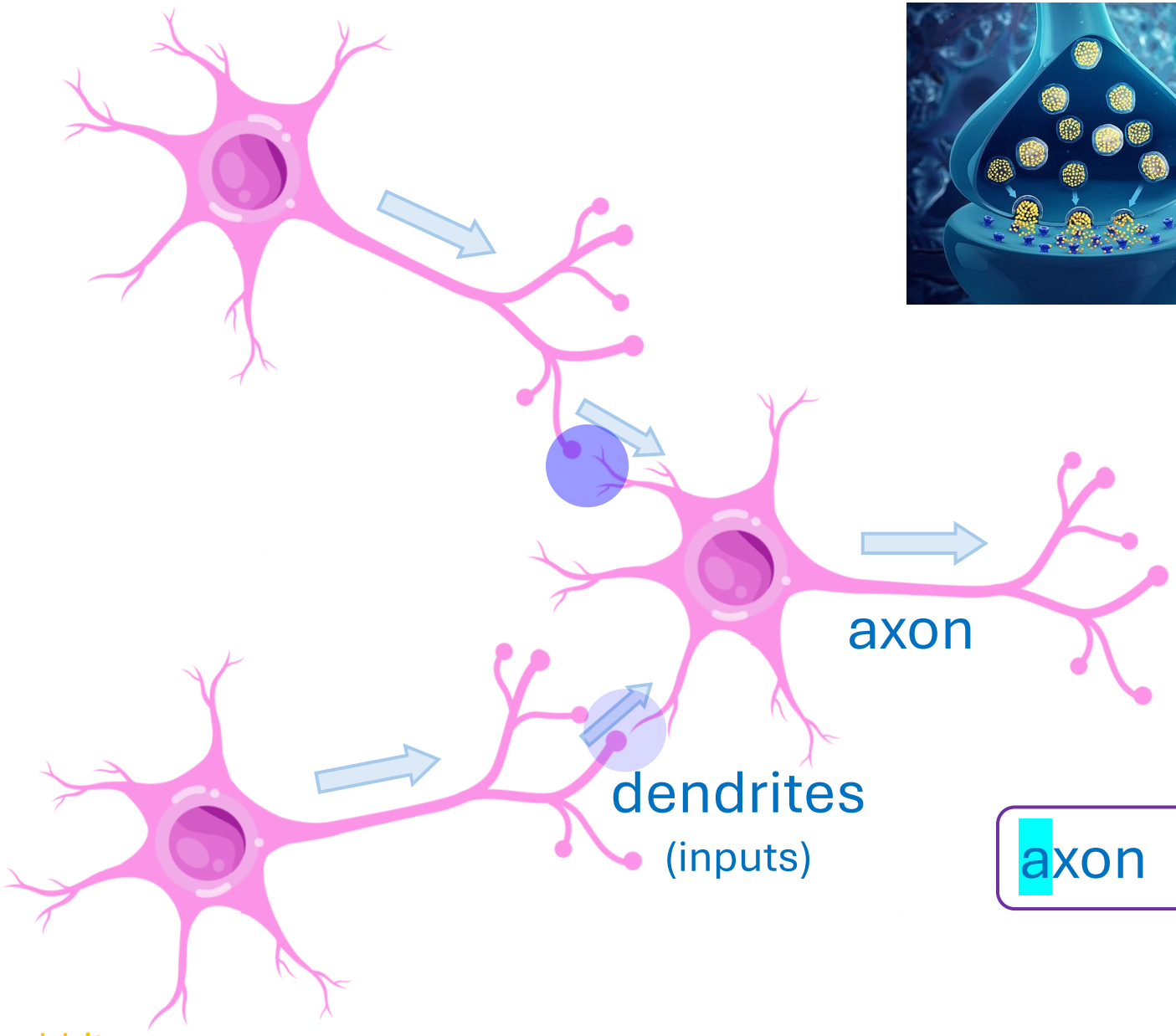




Artificial
Neural Network

Neural Network

Neural Net



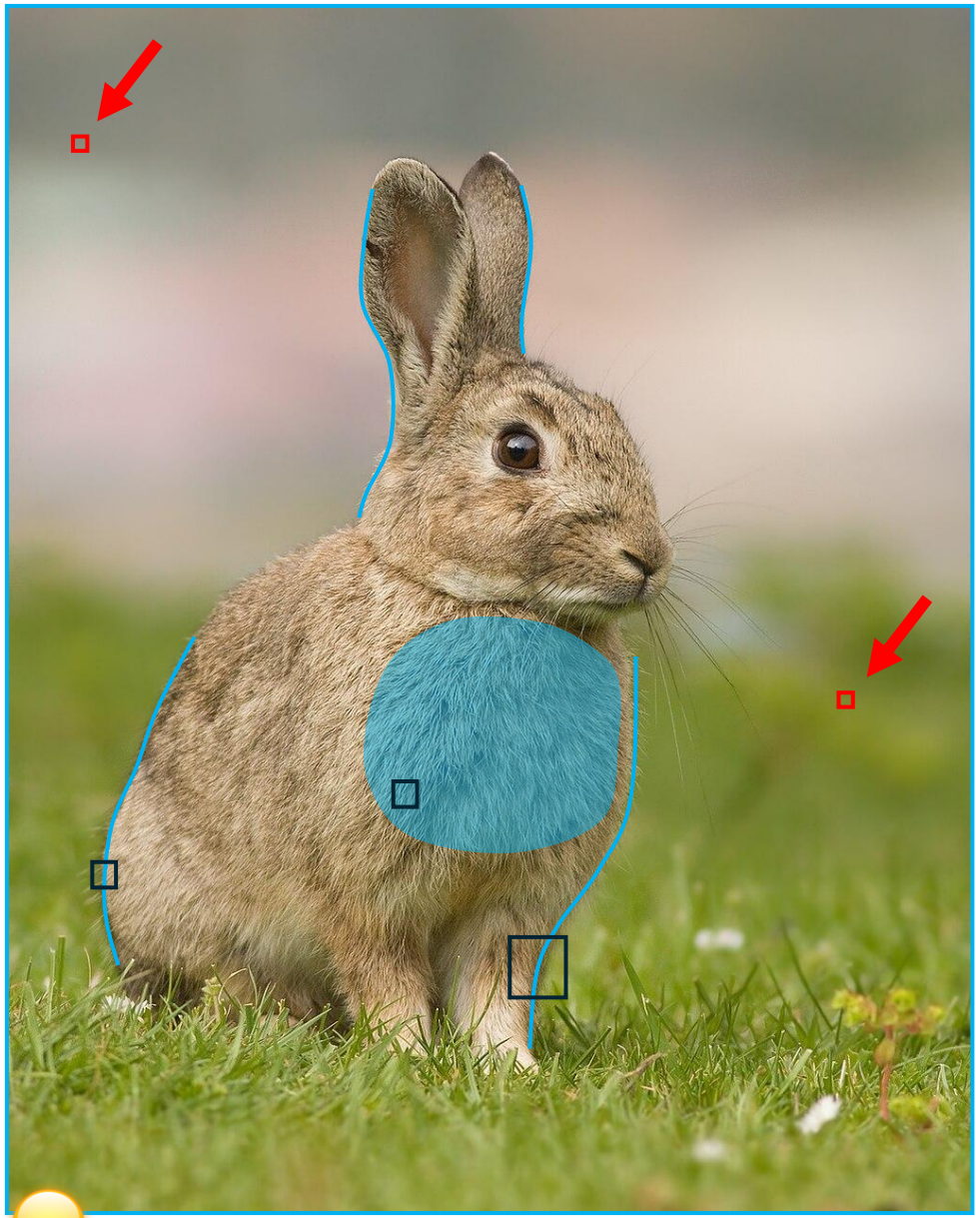
axon

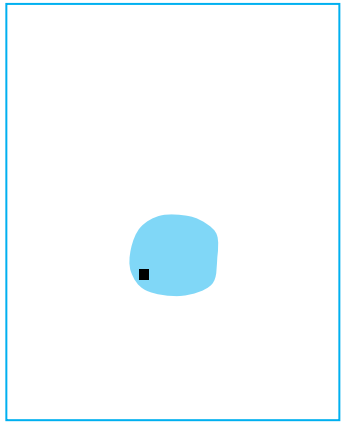
dendrites
(inputs)

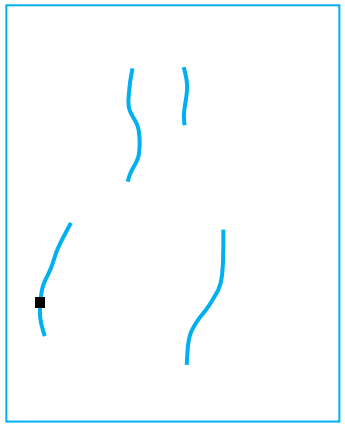
axon away

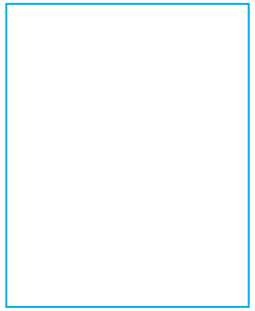
rabbit

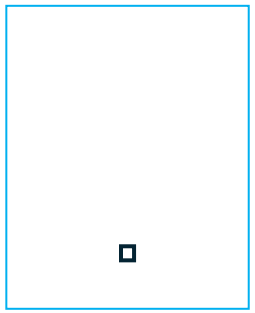
Neuron by Hariadhi / CC BY-SA 4.0
Synapse by www.scientificanimations.com / CC BY-SA 4.0

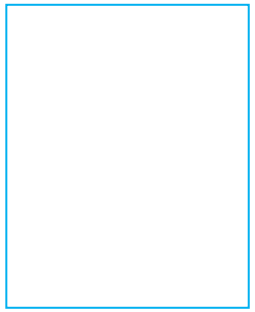




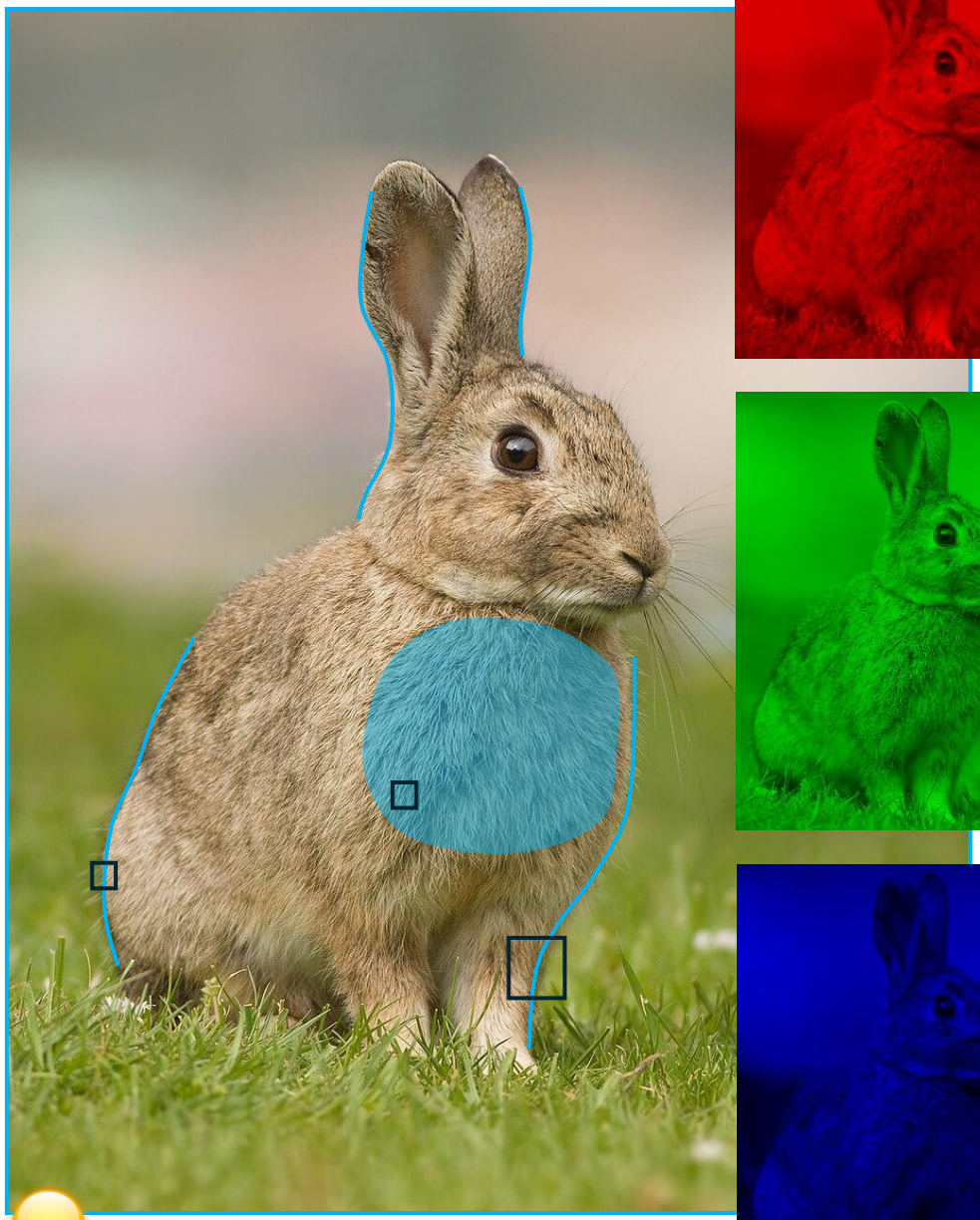








Three colour channels

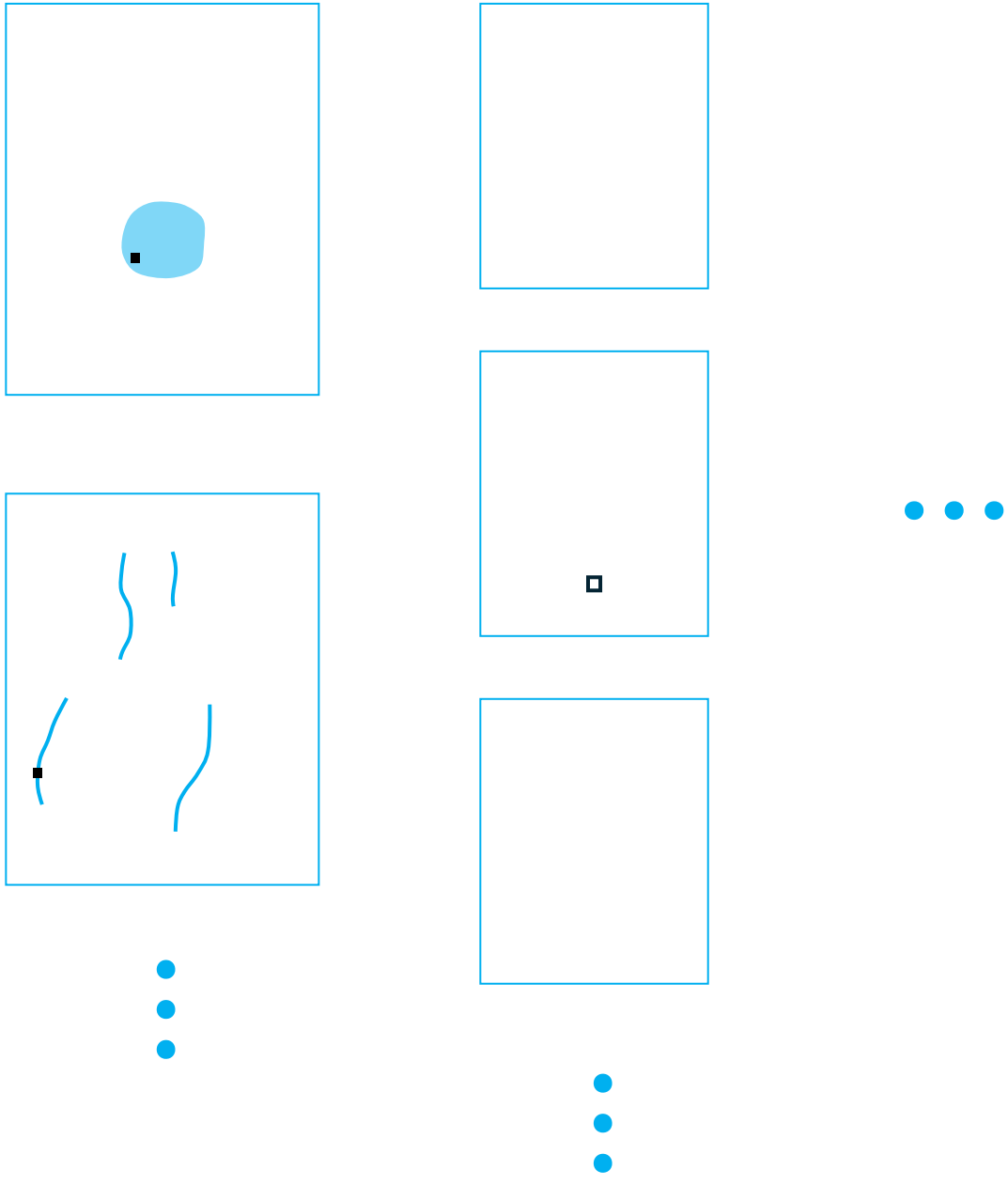


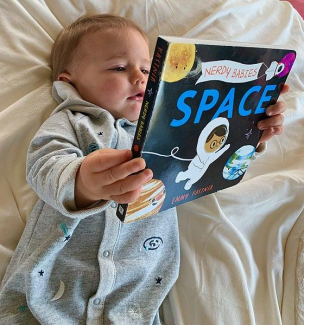






More channels



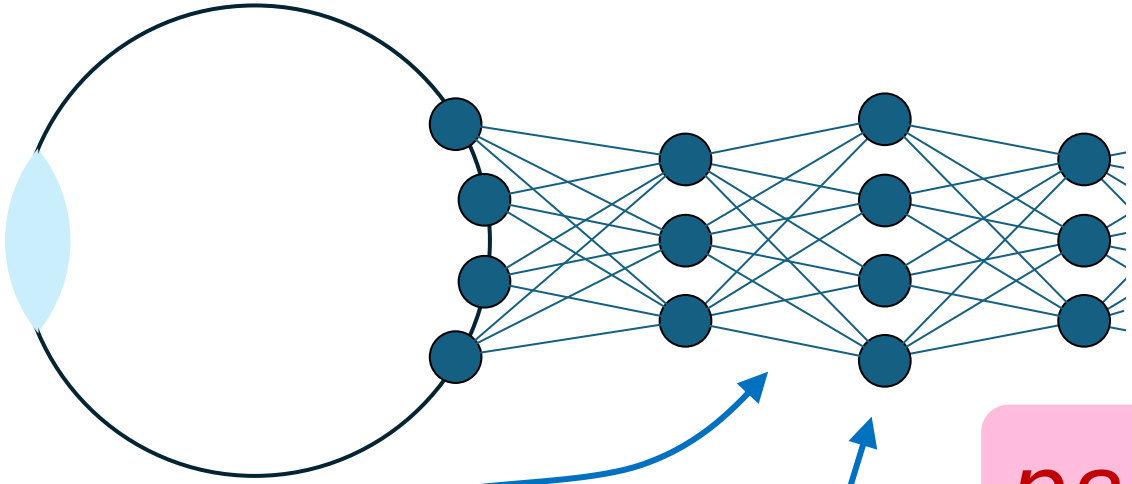


1,000,000,000 ~ billion neurons

1,000,000,000,000 ~ trillion connections

scale

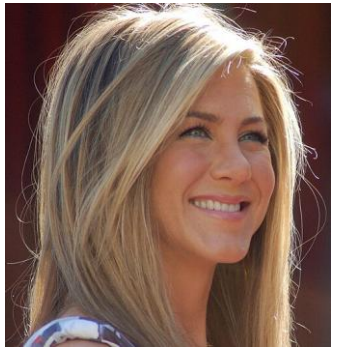




cat

grandmother

Jennifer Aniston



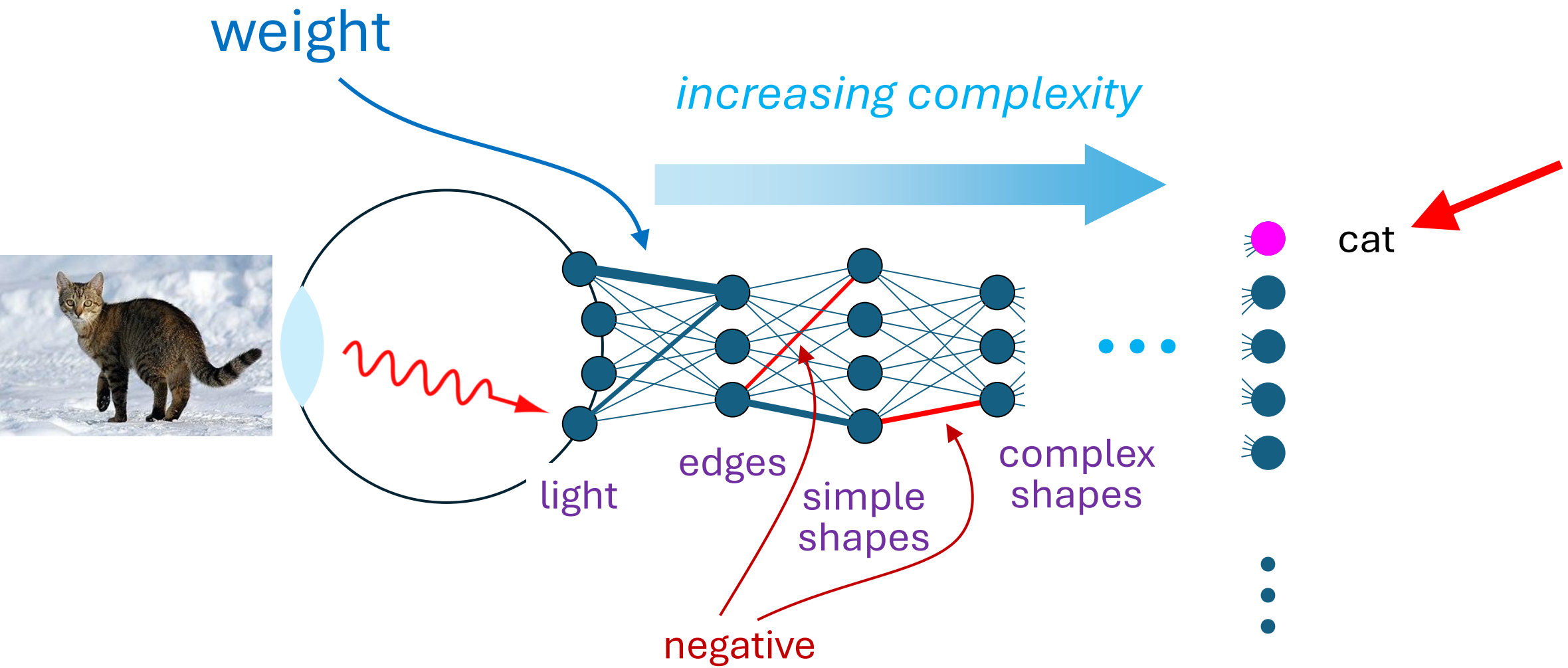
parallel

synapse

slow neuron

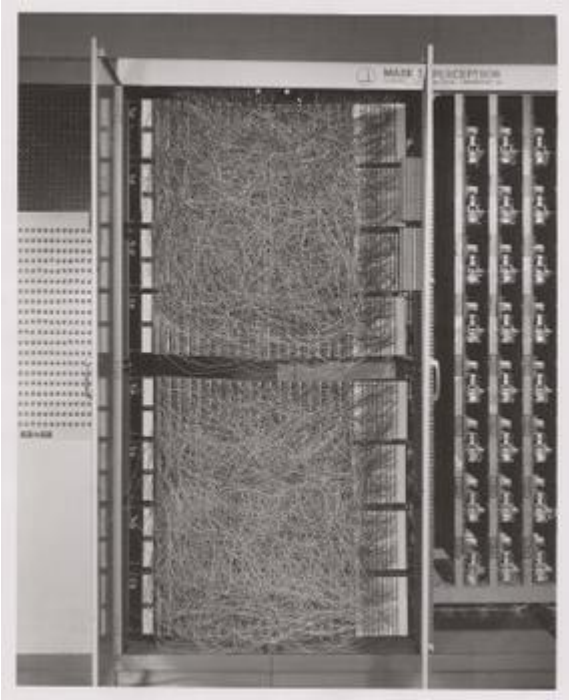
a miniature computer

Babies by Dustin M. Ramsey / CC BY-SA 2.5
Jennifer Aniston by Angela George / CC BY-SA 3.0


weight
increasing complexity
cat
light
edges
simple shapes
complex shapes
negative

perceptron

Mark I Perceptron 1958

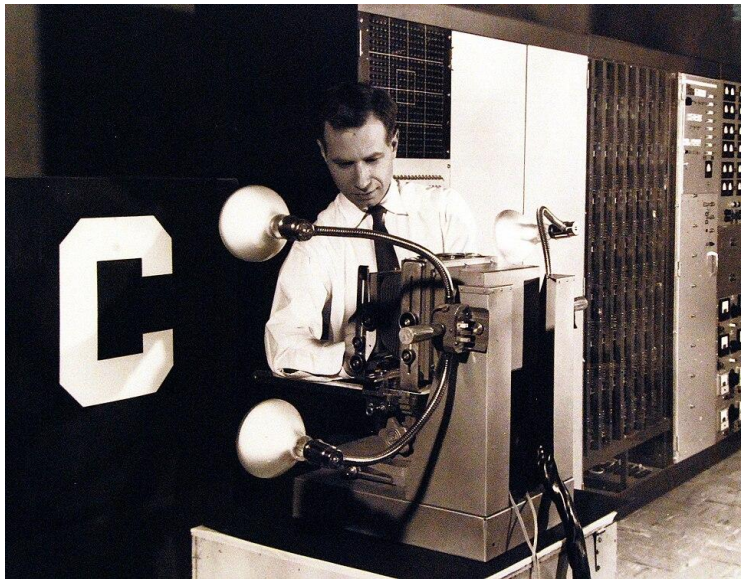


The proposal

"concept formation …

… inductive logic"

The reality



https://www.google.com/search?udm=2&q=mark+i+perceptron

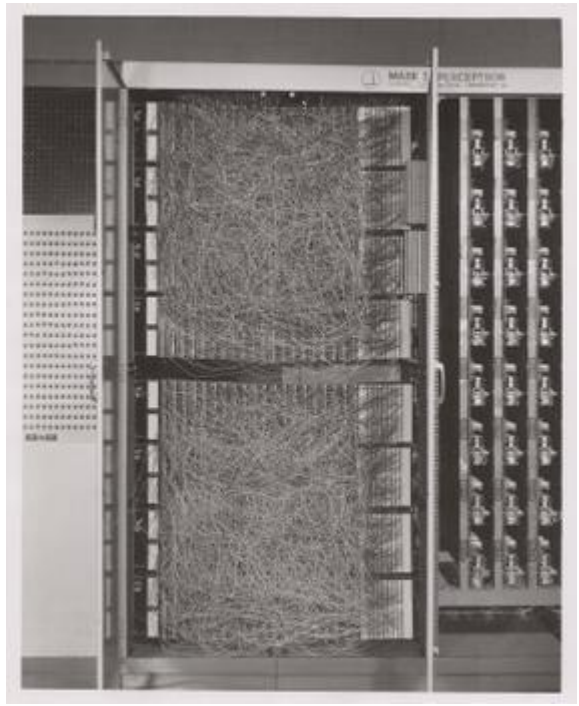
(image search)



?

Perceptron / CC0
Perceptron / Fair use

Mark I Perceptron 1958





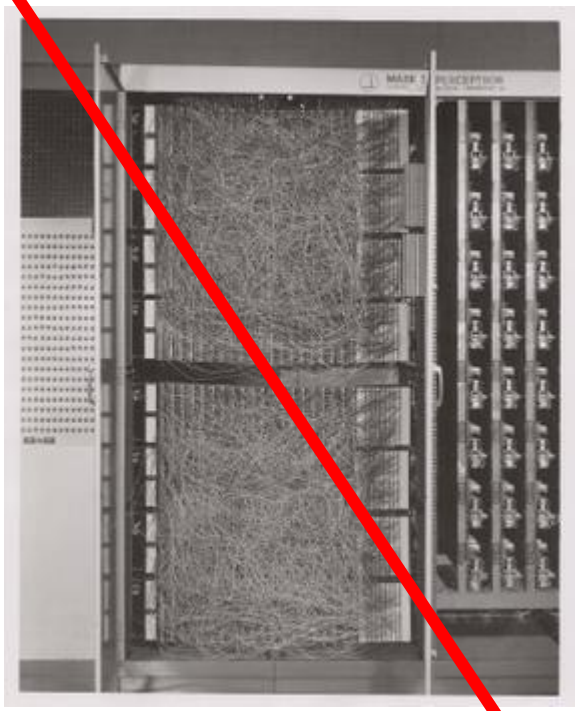




Perceptron / CC0

Perceptron / Fair use

Mark I Perceptron 1958



Physical model

Simulation

(calculations)

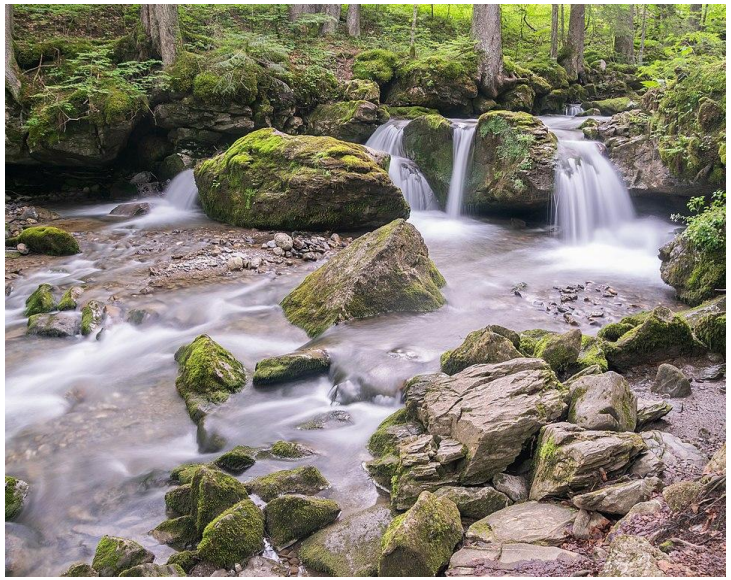
decisions

Perceptron / CC0
Perceptron / Fair use



food ✓

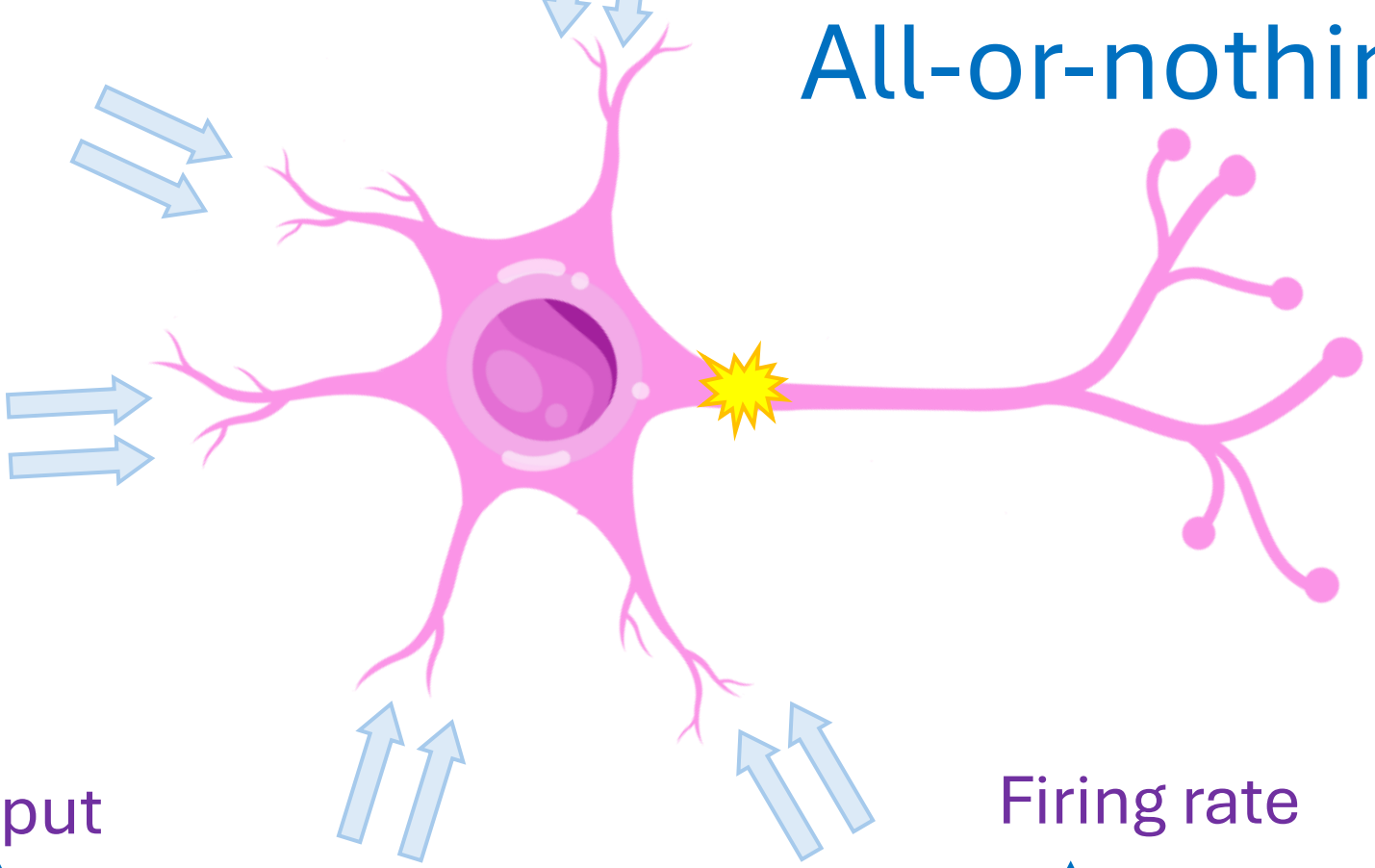
✗



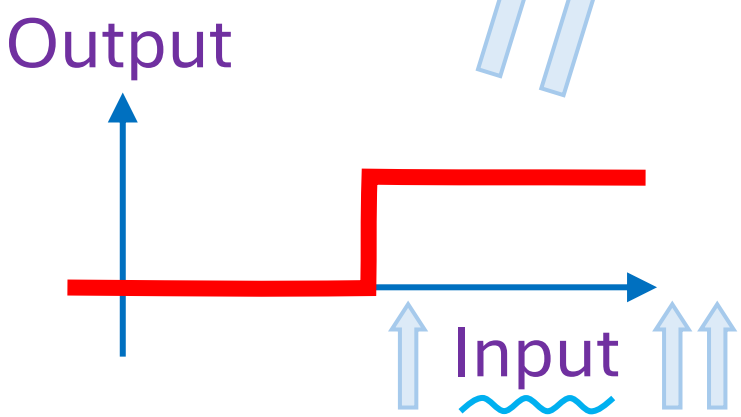
✓

Stream By Krzysztof Golik CC BY-SA 4.0

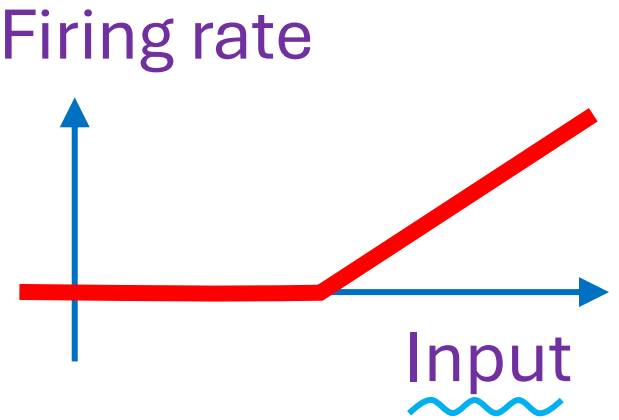
All-or-nothing law





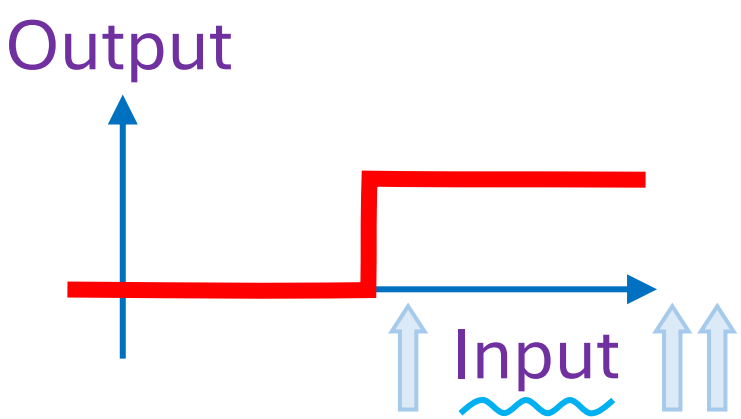


binary ?

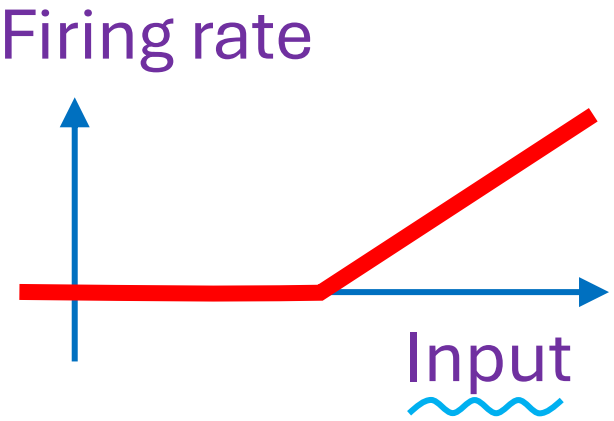


analog ?

Dominos by Kurt:S / CC BY 2.0
Neuron by Hariadhi / CC BY-SA 4.0



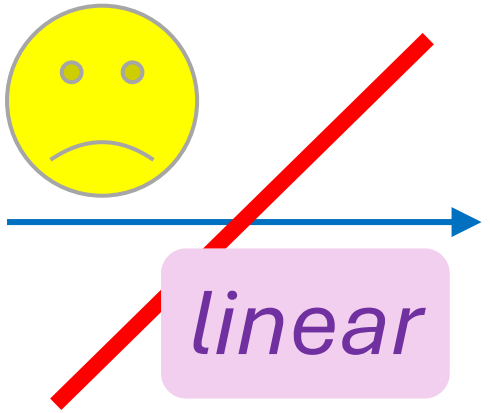
binary ?

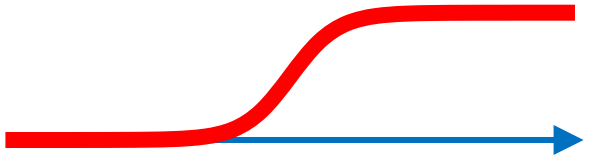


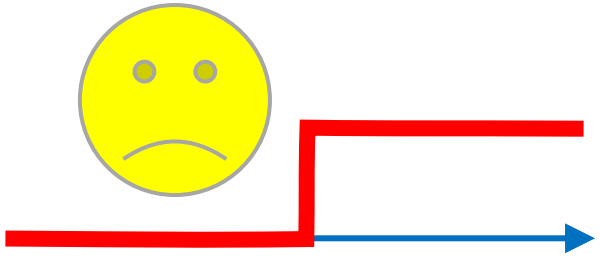
analog ?

Neuron by Hariadhi / CC BY-SA 4.0

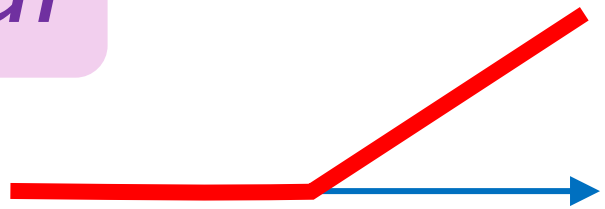
Activation functions







nonlinear



ReLU

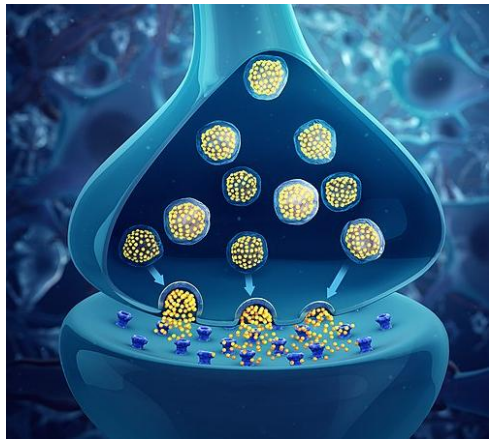
Rectified linear unit

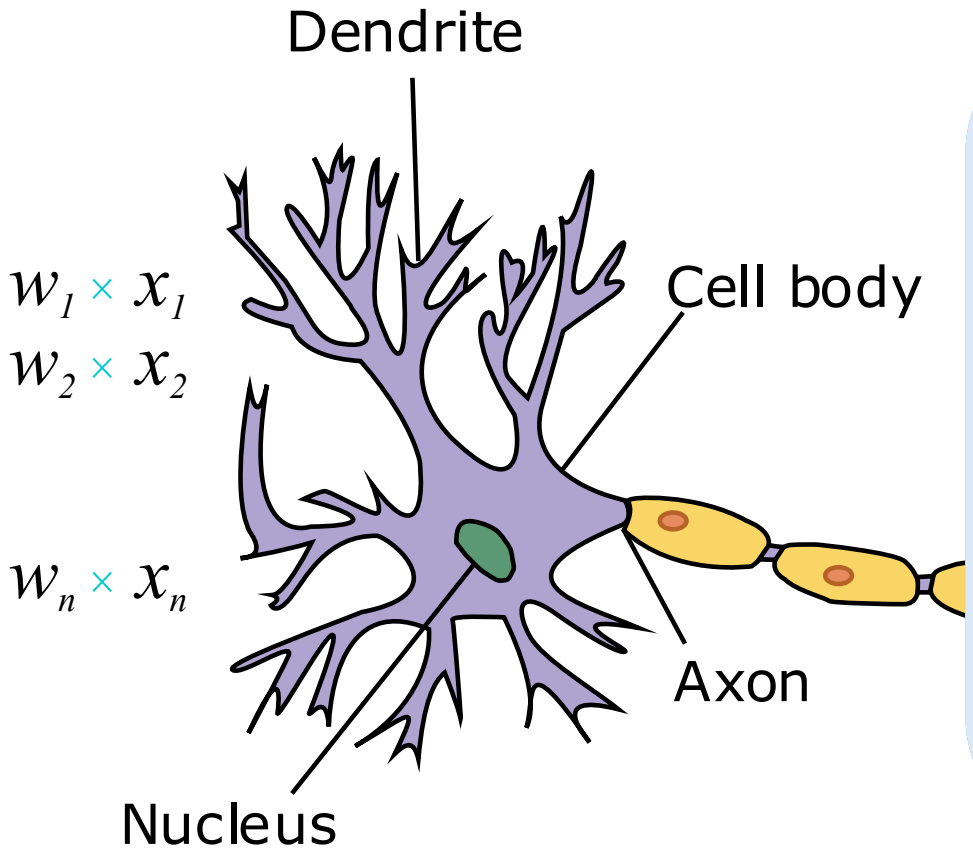


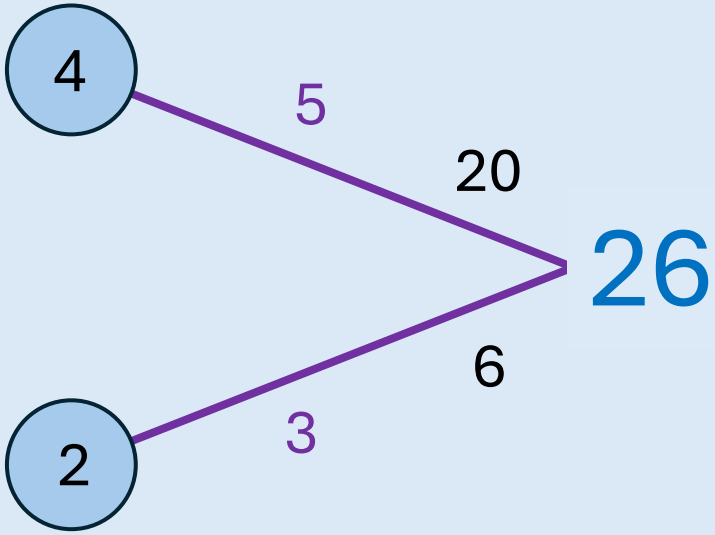
$$x_1 + x_2 + \dots + x_n$$

$$w_1 \times x_1 + w_2 \times x_2 + \dots + w_n \times x_n$$

Synapse





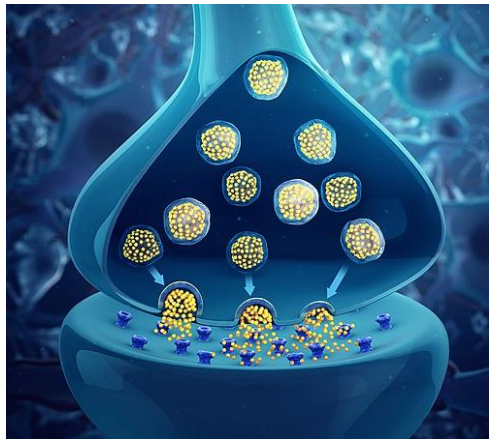


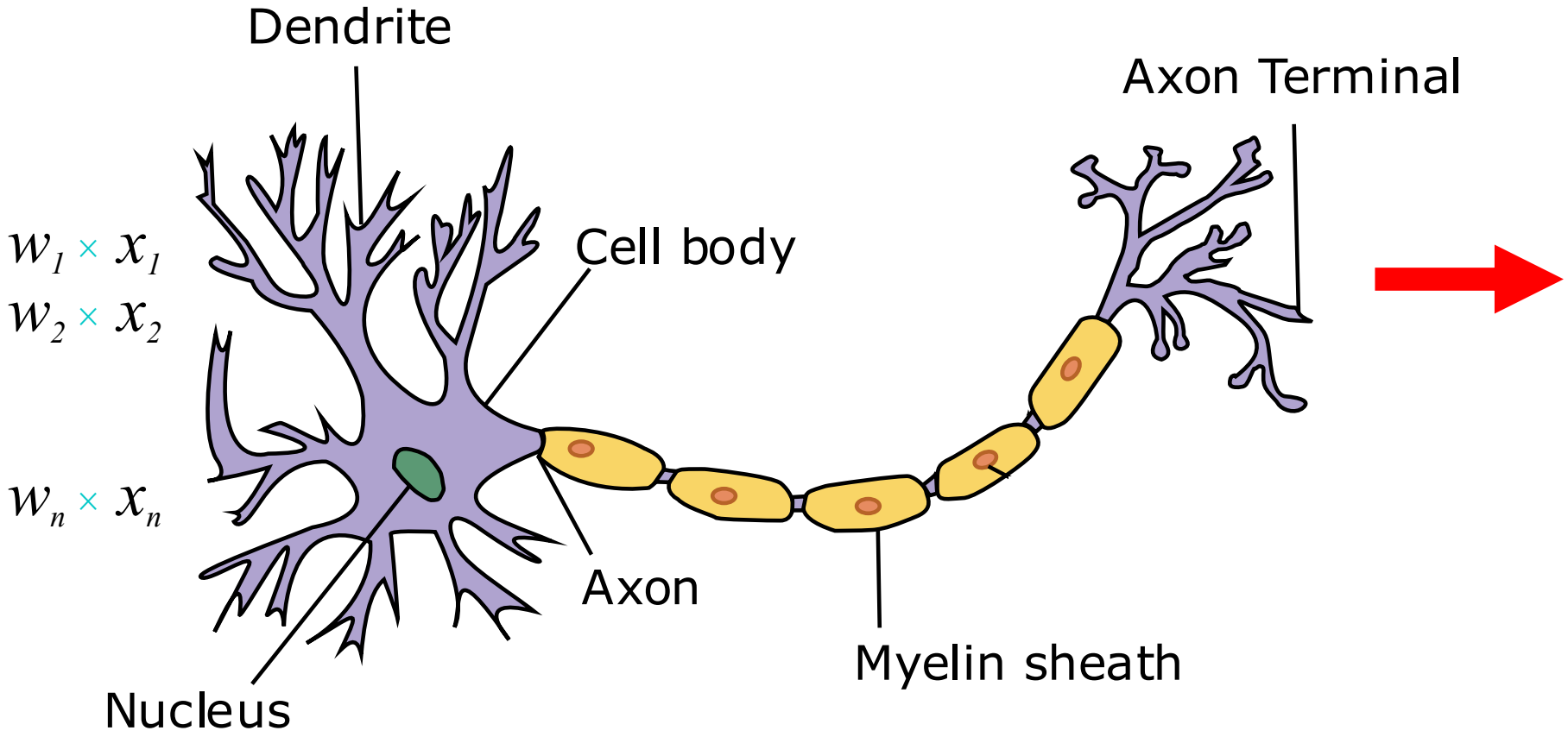
Synapse by www.scientificanimations.com / CC BY-SA 4.0
Neuron by Dhp1080, the US National Cancer Institute's (SEER) Program / CC BY-SA 3.0

$$x_1 + x_2 + \dots + x_n$$

$$w_1 \times x_1 + w_2 \times x_2 + \dots + w_n \times x_n + bias$$

Synapse



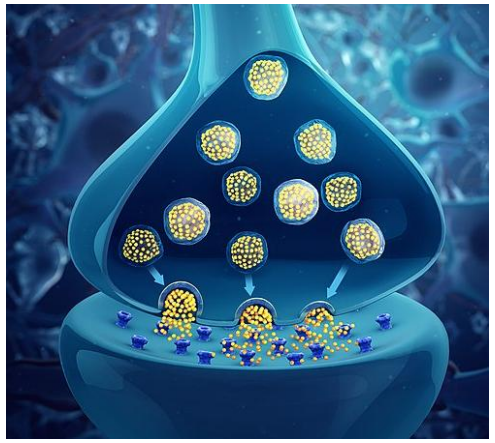


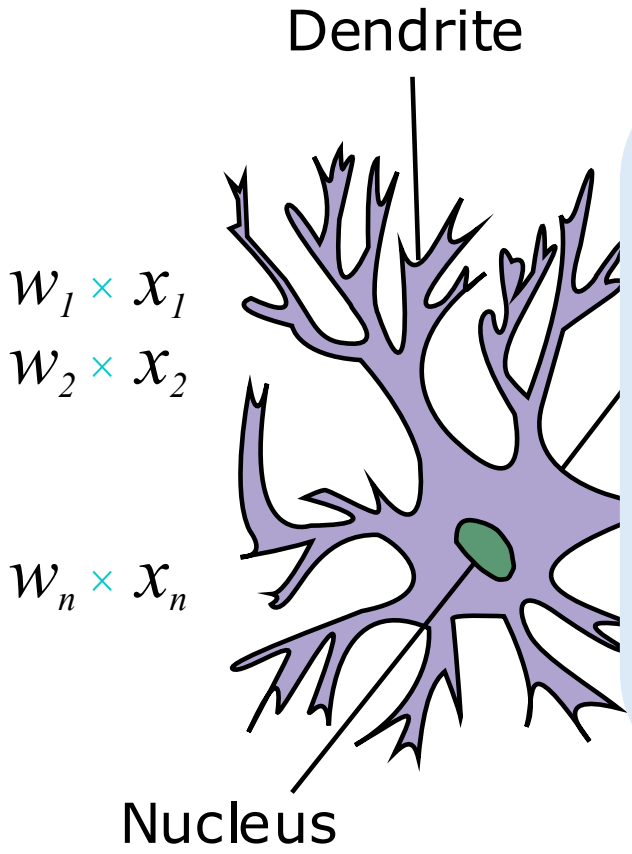
Synapse by www.scientificanimations.com / CC BY-SA 4.0
Neuron by Dhp1080, the US National Cancer Institute's (SEER) Program / CC BY-SA 3.0

$$x_1 + x_2 + \dots + x_n$$

$$w_1 \times x_1 + w_2 \times x_2 + \dots + w_n \times x_n + bias$$

Synapse





weights
biases
} parameters

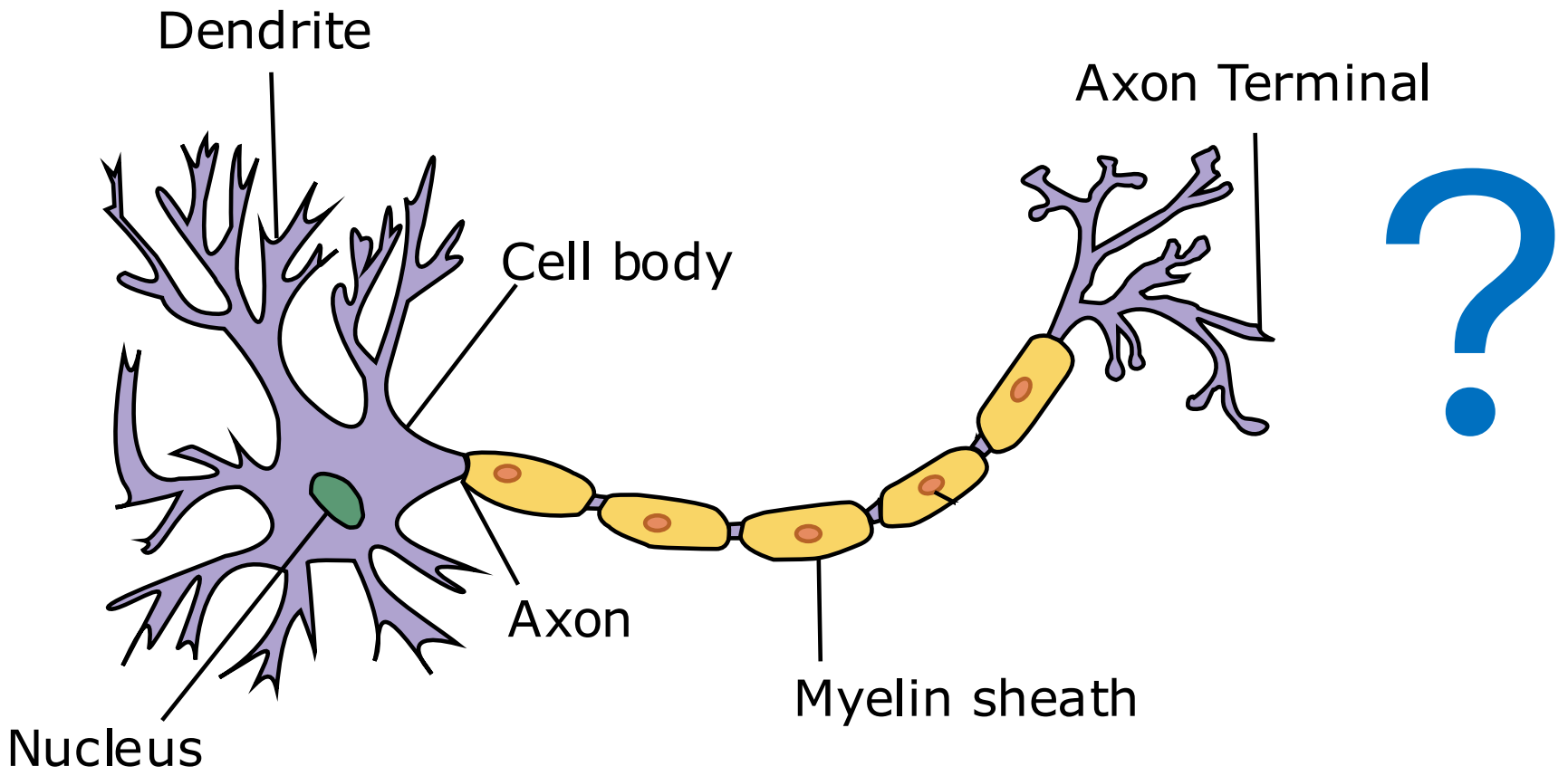
Synapse by www.scientificanimations.com / CC BY-SA 4.0
Neuron by Dhp1080, the US National Cancer Institute's (SEER) Program / CC BY-SA 3.0

Input

$w_1 \times x_1$
$w_2 \times x_2$

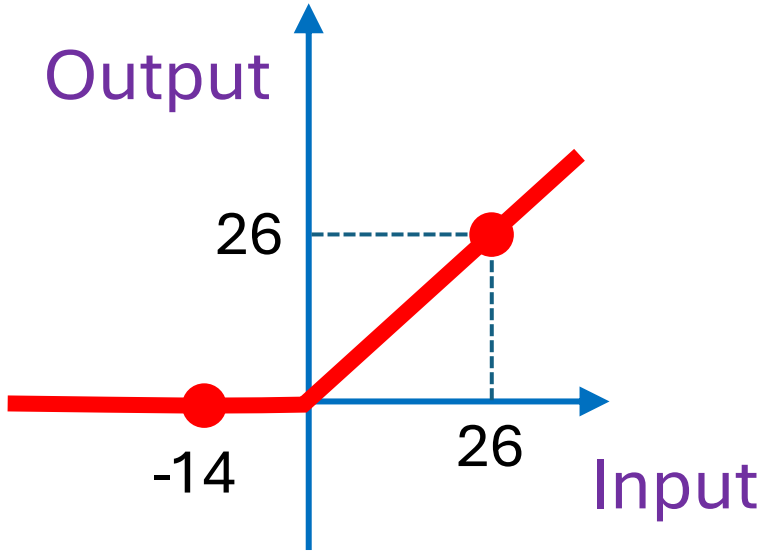
$w_n \times x_n$

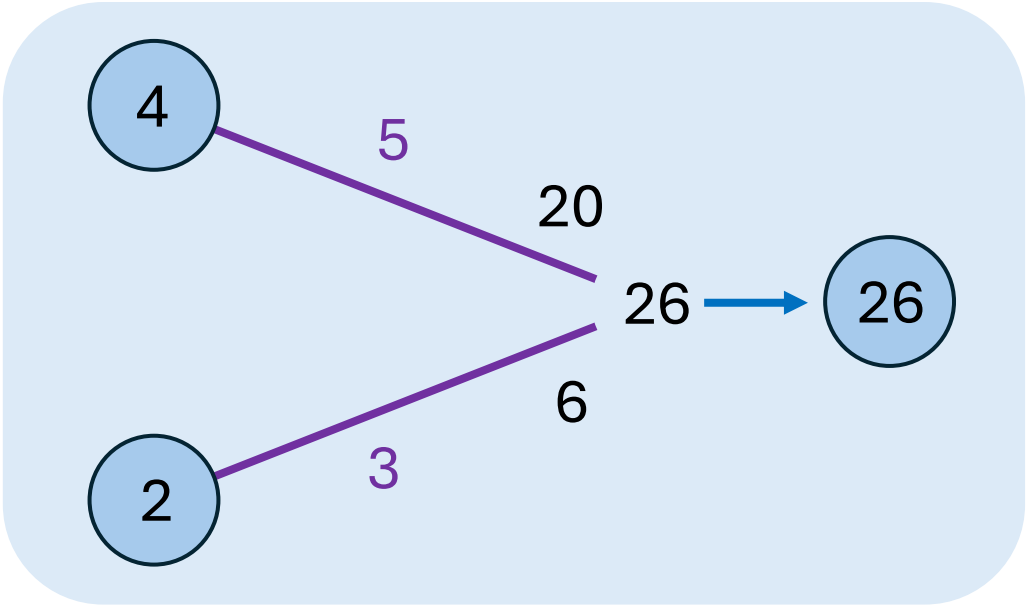
$bias$

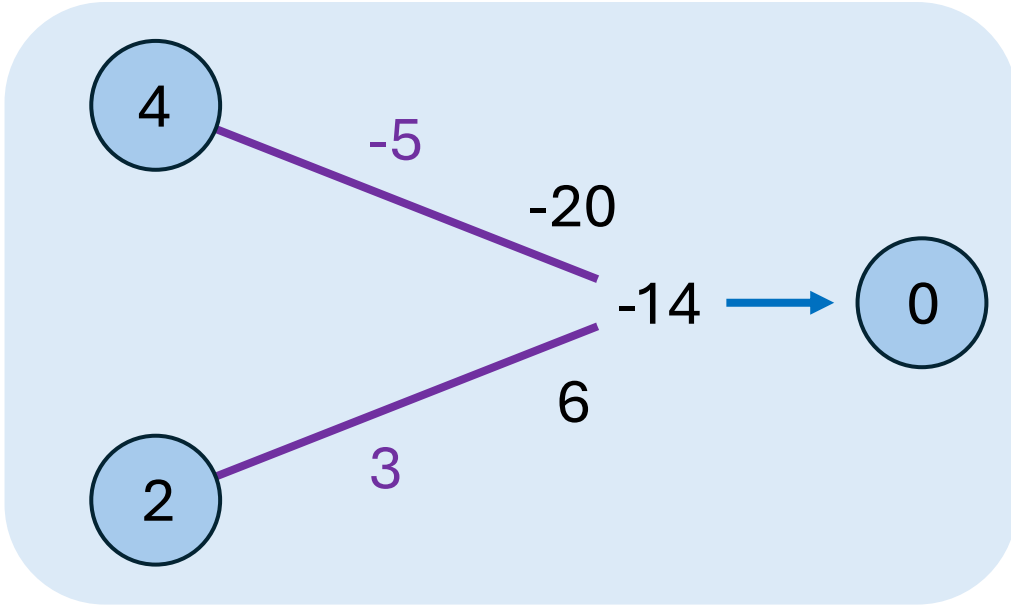


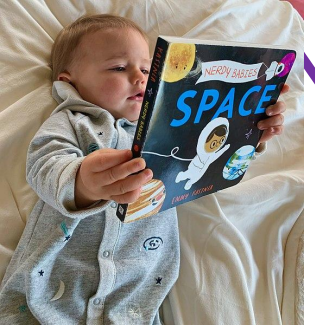
?

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Output
26
-14
26
Input


4
5
20
26
26
6
2
3


4
-5
-20
-14
0
6
2
3



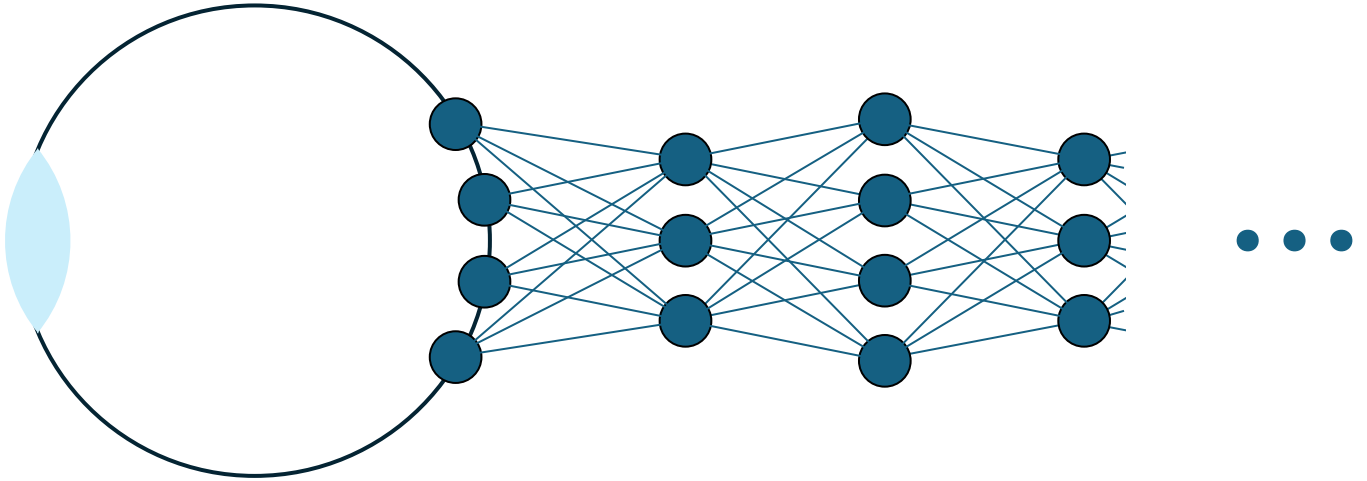
Net



650,000 ~ billion neurons

60 million ~ trillion connections





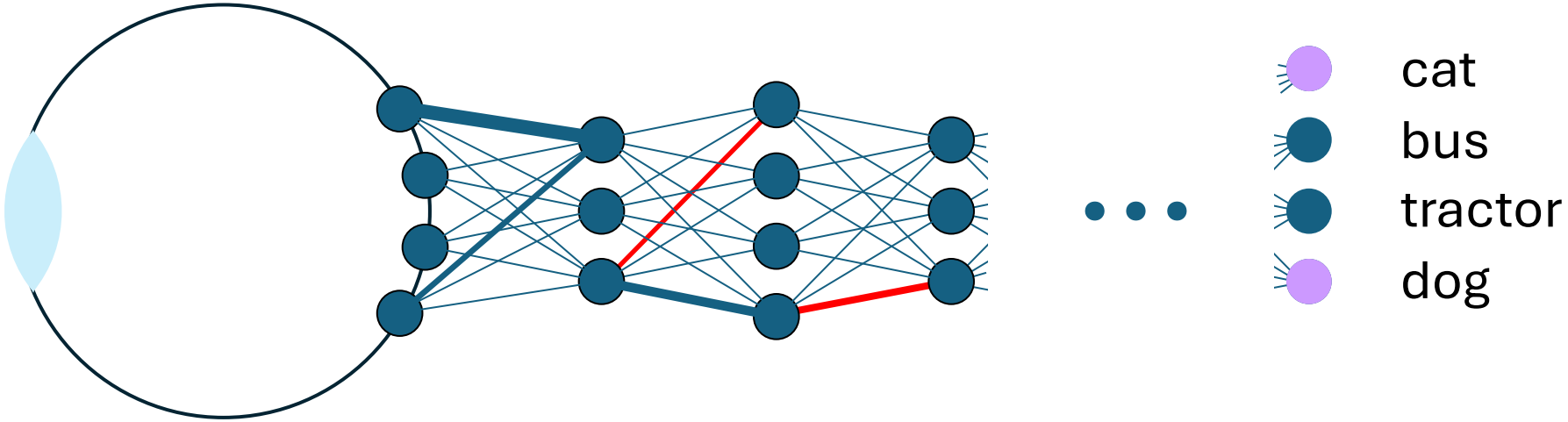
cat



Honey bee by Louise Docker / CC BY 2.0
Babies by Dustin M. Ramsey / CC BY-SA 2.5

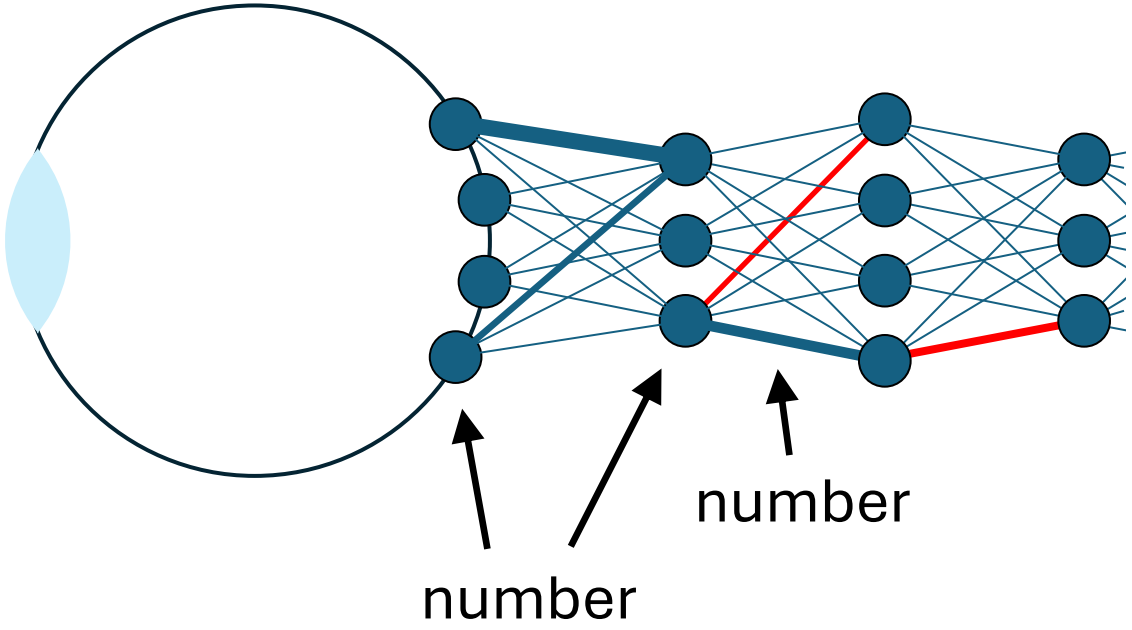
AlexNet

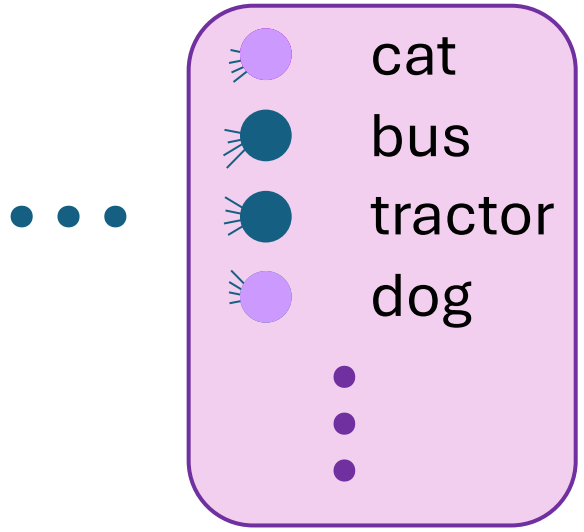




AlexNet







1000 categories

learning

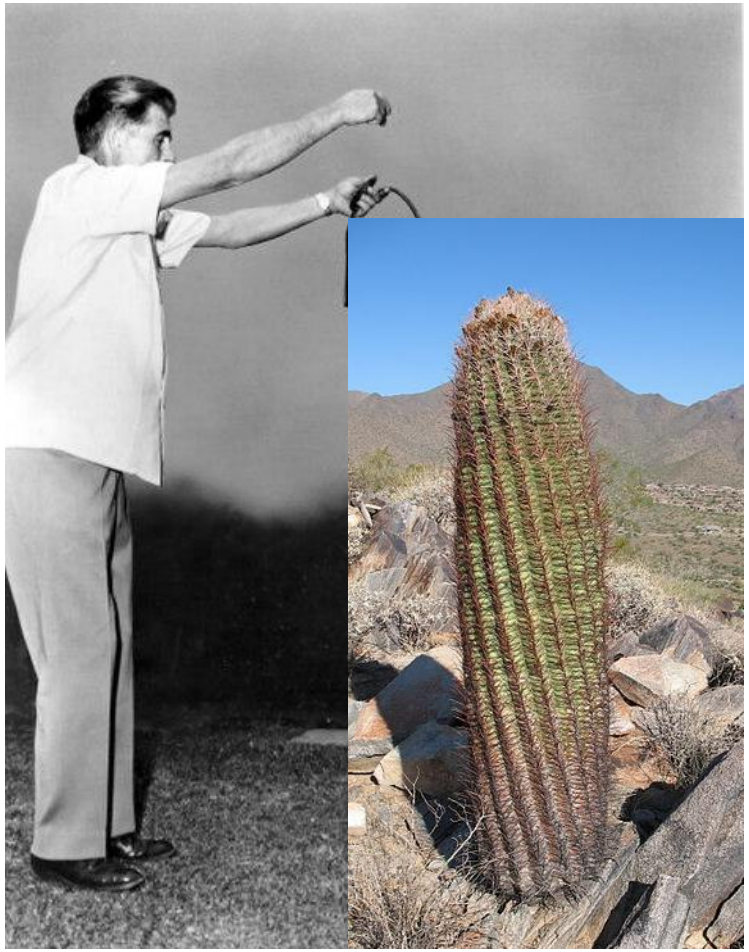
Learning

deeply unobvious



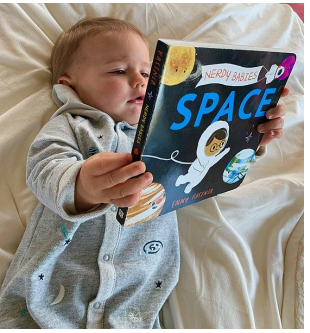
Usual explanation

… feed …
… training data …
… learn



Cactus by Dmcdevit / CC0
Lassie by Batten, Barton, Durstine & Osborn / CC0

How babies learn

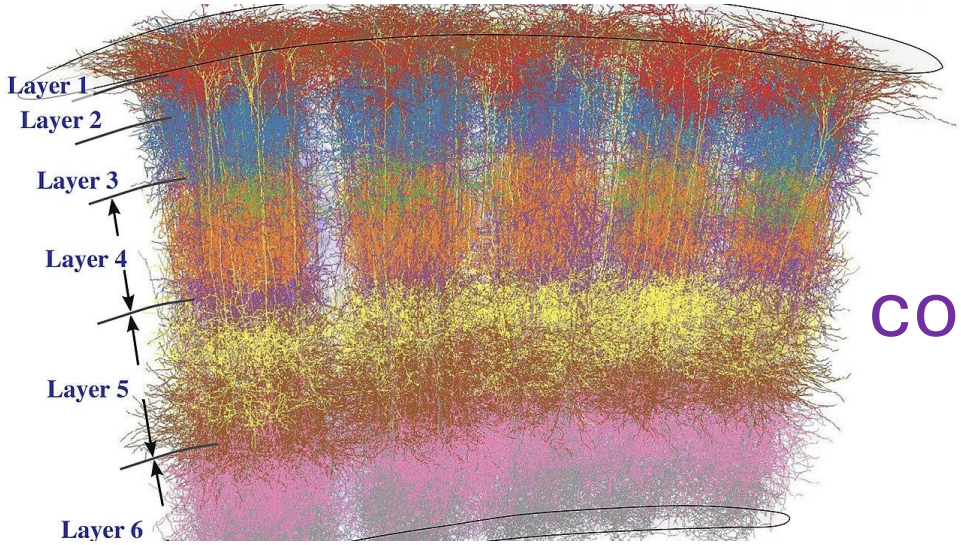


Looking at the world

Exploring



hard work



Making connections

“Neurons that fire together, wire together”

Hebbian theory – Donald Hebb (1949)

“Use it or lose it”

“Stay in sync or lose the link”

A million new connections per second

Neocortex by Bodysurfinyon / CC BY-SA 4.0